Structural Estimation of Directional Dynamic Games with Multiple Equilibria*

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Abstract

We develop a robust algorithm for computing the nested full solution maximum likelihood estimator for a class of directional dynamic stochastic games with multiple equilibria. We show how the computational burden of the full solution approach can be substantially reduced in large datasets, making it computationally feasible. The proposed estimator is remarkably robust to multiplicity of equilibria in the theoretical model, and reliably delivers maximum likelihood estimates of the structural parameters while identifying the equilibria played in the data. Using the dynamic model of Bertrand competition with cost-reducing investments, we run a series of Monte Carlo experiments to explore the performance of our estimator in comparison to the battery of existing estimators for dynamic games. We find that under multiplicity our estimator outperforms all of the existing estimators in terms of accuracy and reliability, while remaining computationally feasible.

Keywords: Structural estimation, dynamic discrete games, multiple equilibria, directional dynamic games, Markov perfect equilibrium, recursive lexicographic search algorithm, MPEC, NFXP, nested recursive lexicographic search algorithm (NRLS)

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1 Introduction

Structural estimation of a dynamic game confronts two unknowns that cannot be separated: the structural parameters and the equilibrium played in the data. At a trial parameter θ the model may admit a set $\mathscr{E}(\theta)$ of Markov perfect equilibria (MPE), each implying a different likelihood, so full-solution maximum likelihood maximizes not a smooth criterion but an upper envelope $\max_{e\in\mathscr{E}(\theta)}\mathscr{L}(\theta,e)$ of the likelihood correspondence. Both the number and the identity of its equilibrium-specific branches $\mathscr{L}(\theta,e)$ change with θ. Branches cross, appear, and vanish; the envelope is nonsmooth or discontinuous; and an algorithm that follows a single equilibrium silently converts its starting value and convergence path into an equilibrium-selection rule. Even if one equilibrium generated the data, multiplicity elsewhere in the parameter space still shapes the likelihood the estimator must maximize.

This paper delivers a computationally feasible full-solution maximum-likelihood estimator for finite-state directional dynamic games: *nested recursive lexicographical search* (NRLS). It combines the nested fixed point logic of Rust (1987) with the recursive lexicographical search (RLS) algorithm of Iskhakov, Rust and Schjerning (2016) (henceforth IRS2016). At every θ visited by the outer maximization, RLS represents the complete set of MPE as a recursively discovered tree, and NRLS maximizes the likelihood over it. Conditional on the stage-game solver finding all stage equilibria, this inner maximization is exact: a branch is discarded only once the algorithm certifies that no completion of it can attain the current likelihood record. The data select among all equilibria the model admits, and no parametric selection rule is imposed.

The distinction between a local equilibrium estimate and the full-solution likelihood is central. Conditional choice probability (CCP) and other two-step methods avoid repeatedly solving the game and are consistent; optimally weighted minimum-distance variants are also efficient (Pesendorfer and Schmidt-Dengler, 2008; Bugni and Bunting, 2021). Their finite-sample performance, however, hinges on a high-dimensional preliminary CCP estimate. Nested pseudo likelihood (NPL) iterates the policy mapping; efficient pseudo likelihood (EPL) replaces that iteration with a locally stable Newton-type update (Aguirregabiria and Mira, 2007; Aguirregabiria and Marcoux, 2021; Dearing and Blevins, 2025). Both attain full-solution efficiency when the equilibrium is unique, yet both can converge to an inferior branch when it is not. Mathematical programming with equilibrium constraints (MPEC) (Su and Judd, 2012; Egesdal, Lai and Su, 2015) recasts maximum likelihood as a constrained optimization problem, but guarantees only local convergence: it certifies an equilibrium, not the likelihood-maximizing equilibrium. What remains unresolved, and what we address, is reliable global selection among equilibrium branches when multiplicity is empirically consequential.

Directionality is the foundation. Following IRS2016, we give a primitive, one-step characterization: after a common ordering of substages, every conditional state-transition matrix is upper block-triangular—equivalently, the graph of feasible transitions between substages is acyclic. Directionality is therefore verifiable directly from the transition primitives, and a topological sort supplies an order for backward solution. The characterization also delimits the scope of RLS and establishes that its decomposition results survive under the one-step definition used here.

Our first contribution turns brute-force equilibrium enumeration into a certifiable discrete optimization problem. Directionality decomposes the game into a sequence of smaller stage games; choosing one stage equilibrium at each step defines an MPE, and all such recursive choices form a tree whose root-to-leaf paths are exactly the MPE of the game. RLS discovers this tree while traversing it, reusing the computations shared by equilibria with a common prefix. NRLS fuses the traversal with branch and bound (Land and Doig, 1960): at any partially constructed branch, the likelihood accumulated over states already solved is combined with an upper bound on the fit attainable at states not yet reached. If the sum falls below the likelihood of a completed branch, the entire subtree is pruned without solving a single further stage game. The nonparametric likelihood is a valid and tight bound on the remaining contribution of any completion, so exact full-solution estimation does not require full enumeration.

The same argument yields a counterintuitive computational implication. As the number of independent markets pooled under an equilibrium grows, empirical choice frequencies become more informative about which branches can fit the data: the normalized gap between the nonparametric bound and the likelihood attainable by the data-generating equilibrium vanishes, while population likelihood differences between separated equilibria persist. Inferior branches are then rejected closer to the root of the tree. More observations raise the cost of processing the data but cut the far larger cost of solving stage games and exploring equilibrium branches. In this precise sense information is a computational resource: for NRLS, more data can mean fewer computations.

Our second contribution is an econometric theory for maximizing a parameter-dependent likelihood correspondence. Let a common equilibrium generate independent markets, each observed for a fixed number of periods. Under joint identification and population separation of the structural parameter and the equilibrium, NRLS consistently estimates both. The probability of selecting the data-generating equilibrium converges to one, and the parameter estimator is first-order equivalent to the oracle maximum-likelihood estimator that knows that equilibrium. When the true parameter is interior to the equilibrium's existence region, the estimator is asymptotically normal at the usual root-M rate. Because the asymptotics increase the number of markets while holding their histories fixed, no stationarity of the within-market state process is required—an essential feature for directional games, whose states eventually reach an absorbing region.

Multiplicity also generates sampling uncertainty that smooth-likelihood theory does not describe. When the true parameter lies on the boundary of the existence region of the data-generating equilibrium, the branch supporting that equilibrium disappears on one side, and the limit is nonstandard: in the scalar one-sided case, a one-sided Gaussian component plus an atom at the boundary (Andrews, 1999). Near such a boundary finite-sample distributions can look sharply truncated even though a fixed interior sequence eventually returns to the oracle normal approximation. The likelihood correspondence governs inference, not just computation.

Our contribution complements several strands of the literature. Work on identification and estimation under multiplicity has emphasized that equilibrium selection, unobserved heterogeneity, and the cardinality of the equilibrium set cannot generally be treated independently (de Paula, 2013; Aguirregabiria and Mira, 2019); approaches robust to an unspecified selection mechanism typically report identified sets or moment-inequality conclusions rather than a point estimate (Otsu, Pesendorfer, Takahashi and Sasaki, 2022). Earlier procedures search over equilibrium assignments with genetic algorithms or Bayesian simulation (Aguirregabiria and Mira, 2005; Misra, 2013), while constrained optimization methods formulate equilibrium and estimation jointly (Su, 2014; Su and Judd, 2012; Egesdal, Lai and Su, 2015). NRLS takes a different route: it exploits the recursive structure of directional dynamic games to search the complete MPE set and supplies a deterministic certificate for the inner likelihood maximum. Multiplicity and discontinuous likelihoods are not new observations; what is new is that in this class of dynamic games their global likelihood consequences can be computed exactly without enumerating the equilibrium set.

Our third contribution is a systematic computational and statistical stress test. We use the dynamic Bertrand investment game of Iskhakov, Rust and Schjerning (2018) (henceforth IRS2018), a parsimonious model in which duopolists make cost-reducing investments and can leapfrog one another. Small changes in its primitives generate a unique equilibrium, a few stable or unstable equilibria, discontinuities where an equilibrium branch ceases to exist, or thousands of MPE. This variation lets us display the equilibrium-specific likelihood curves and their upper envelope rather than treat the objective as a black box. We compare NRLS with one-step and iterated NPL (Aguirregabiria and Mira, 2007) (henceforth AM2007), one-step and iterated EPL (Dearing and Blevins, 2025), and two MPEC formulations (Su, 2014; Su and Judd, 2012; Egesdal, Lai and Su, 2015).

The experiments separate local numerical success from global statistical reliability. With a unique equilibrium, the equilibrium-imposing estimators largely agree. Under multiplicity, NPL can fail to converge, while EPL and MPEC can converge with essentially zero equilibrium residual and still sit on a likelihood-inferior branch. A dedicated convergence experiment shows the NRLS estimates approaching the regular root-M approximation under a smooth likelihood, while the design close to an equilibrium-existence boundary exhibits the pronounced truncation predicted by theory. Meanwhile equilibrium misclassification disappears rapidly and branch and bound explores fewer branches as the number of markets grows: the statistical and computational returns to information reinforce one another.

The scale experiments make the main computational result concrete. In a game with 2,455 MPE, NRLS estimation takes 4.73 seconds, whereas enumerating the equilibrium set once takes over 10 minutes. In our largest reported design the model has 19,683 MPE and 25 prespecified market groups play 23 distinct equilibria; NRLS recovers all 25 assignments in every Monte Carlo replication and averages 189 seconds, while exhaustive RLS enumeration at a single parameter value would take approximately 2.5 hours on the same machine—and a brute-force full-solution estimator would have to repeat that enumeration at every trial parameter. NRLS estimates the model in substantially less time than it takes to solve the model exhaustively even once.

The remainder of the paper proceeds as follows. Section 2 sets up the dynamic game, characterizes directionality, and introduces the Bertrand investment model. Section 3 defines the full-solution likelihood, establishes the statistical results, represents RLS as a tree traversal, and develops the branch-and-bound implementation. Section 4 presents the competing estimators and Monte Carlo evidence. Section 5 concludes.

2 Directional dynamic games

Our basic framework is a discrete version of stochastic game due to Shapley (1953) where some of the state variables constitute private information of the players as in AM2007. The Markov perfect equilibria of these games of incomplete information are characterized by a collection of choice probabilities defined over the common state variables, which allows construction of the likelihood function of the observed sample of actions and states. We introduce additional assumptions to give new characterization of the class of *directional dynamic games* (DDG) considered in IRS2016. The dynamic Bertrand pricing and investment game from IRS2018 is a running example that we return to throughout the paper.

2.1 Discrete dynamic games of incomplete information

We follow AM2007 to set up the canonical dynamic discrete game of incomplete information. Consider a stochastic game of $N < \infty$ players who in each time period choose an action $a_i \in A$ from a finite set of actions $A = \{0, \dots, J\}$. Vector $a = (a_1, \dots, a_N)$ is the action profile of all players in the current time period. The state of the game is given by a finite set of variables x_i which are known to all players, and a set of private ε_i known only to player i. The vectors $x = (x_1, \ldots, x_N)$ and $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_N)$ are the common knowledge and private information, respectively, where $x \in X$ has a discrete and finite support X, $|X| < \infty$. We assume that (x, ε) follows a controlled Markov process with transition probability $p(x', \varepsilon'|a, x, \varepsilon)$, where (x', ε') refers to the next period values. The instantaneous payoff of player i is given by $\tilde{\Pi}_i(a, x, \varepsilon_i)$. We further impose the following standard assumptions.

Assumption A (model primitives)**.** *(AS) Additive separability: the per-period payoff of player i is additively separable in x and ε_i, i.e. $\tilde{\Pi}_i(a,x,\varepsilon_i) = \Pi_i(a,x) + \eta \varepsilon_i[a_i]$, where $\varepsilon_i[a_i]$ is the (a_i+1)-th component of vector $\varepsilon_i \in \mathbb{R}^{J+1}$ and η is scaling parameter; (CI) Conditional independence: The state (x, ε) evolves as a first-order controlled Markov process with transition density $p(x', \varepsilon'|a, x, \varepsilon) = f(x'|a,x) \cdot p_\varepsilon(\varepsilon')$, where f governs the transition of the common knowledge state and p_ε is the time-invariant distribution of private shocks; (IPV) Independent private values: Private shocks are independently distributed across players, i.e. $p_\varepsilon(\varepsilon) = \prod_{i=1}^N g_i(\varepsilon_i)$ where g_i is the probability density function of ε_i; (EV) Extreme value[1]: Each private shock ε_i is distributed according to the extreme value type I (Gumbel) distribution, i.e. $g_i(\varepsilon_i) = \exp(-\varepsilon_i - \exp(-\varepsilon_i))$.*

Let $\sigma = (\sigma_1, \ldots, \sigma_N)$ denote a Markovian strategy profile, where $\sigma_i : X \times \mathbb{R}^{J+1} \to A$ is a decision rule mapping of each player. The behavior of each player under strategy profile σ is perceived by the other players through the choice probabilities denoted $P_i^\sigma(a_i|x) \equiv Pr\{\sigma_i(x, \varepsilon_i) = a_i|x\}$. Denote $a_{-i} \in A^{N-1}$ the vector of all possible actions of the players other than player i. Under a strategy profile σ the deterministic part of i's instantaneous payoff as well as the transition probability for the common knowledge states seen from i's point of view can be expressed as a function of their action a_i as

$$\begin{aligned} \pi_i^\sigma(a_i,x) &= \textstyle\sum_{a_{-i}\in A^{N-1}} \left(\prod_{j\neq i} P_j^\sigma\left(a_{-i}[j]\middle|x\right)\right) \Pi_i\big((a_i,a_{-i}),x\big), \\ f_i^\sigma(x'|a_i,x) &= \textstyle\sum_{a_{-i}\in A^{N-1}} \left(\prod_{j\neq i} P_j^\sigma\left(a_{-i}[j]\middle|x\right)\right) f\big(x'|(a_i,a_{-i}),x\big), \end{aligned} \tag{1}$$

where $a_{-i}[j]$ is the j-th element of a_{-i}. The dynamic behavior of player i is then described by the *integrated* Bellman equation

$$V_i^\sigma(x) = \int \max_{a_i \in A} \{v_i^\sigma(a_i,x) + \eta \varepsilon_i[a_i]\}\, g_i(d\varepsilon_i) = \eta \log \sum_{a_i \in A} \exp\left(\frac{v_i^\sigma(a_i,x)}{\eta}\right) + \eta\gamma, \tag{2}$$

where the last equality is due to assumption (EV), $\gamma \approx 0.57721$ is the Euler-Mascheroni constant, and $v_i^\sigma(a_i,x)$ denotes the a_i-choice specific value function of player i. With the common discount factor β

[1]We introduce the EV assumption right away to simplify exposition; all theoretical results and representations follow through with any g_i that are absolutely continuous with respect to Lebesgue measure (Hotz and Miller (1993), AM2007).

it is given by

$$v_i^\sigma(a_i,x) = \pi_i^\sigma(a_i,x) + \beta \sum_{x' \in X} V_i^\sigma(x') f_i^\sigma(x'|a_i,x). \tag{3}$$

Conditional on a fixed strategy profile σ, equations (2) and (3) define a Bellman operator $\Phi : V^\sigma \mapsto V^\sigma$ which is a contraction mapping in the space of value functions $V_i^\sigma(x)$ (Aguirregabiria and Mira, 2002; Ma and Stachurski, 2021).

Definition 1. *A strategy profile $\sigma^\star$ constitutes a stationary Markov perfect equilibrium (MPE) of the game if for any player i and for any state $(x,\varepsilon_i) \in X \times \mathbb{R}^{J+1}$ it holds*

$$\sigma_i^\star(x,\varepsilon_i) = \arg\max_{a_i \in A} \left\{ v_i^{\sigma^\star}(a_i,x) + \eta \varepsilon_i[a_i] \right\}, \tag{4}$$

that is the strategy profile $\sigma^\star$ prescribes mutual best responses that solve Bellman equation (2) for $i \in \{1,\dots,N\}$.

The MPE equilibrium can equivalently be characterized in probability space. First, note that all payoff relevant information, namely π_i^σ, f_i^σ, V_i^σ and v_i^σ depend on players' strategies only through the choice probabilities $P_i^\sigma(a_i|x)$ associated with σ. Given (EV), the choice probabilities take logit form

$$P_i^\sigma(a_i|x) = Pr\{\sigma_i(x,\varepsilon_i) = a_i|x\} = \frac{\exp(v_i^\sigma(a_i,x)/\eta)}{\sum_{a_i' \in A} \exp(v_i^\sigma(a_i',x)/\eta)}, i \in \{1,\dots,N\}. \tag{5}$$

Denote Λ: $P \mapsto P$ a mapping from the space of choice probabilities defined for all (a,x) for all players $i \in \{1,\dots,N\}$, to itself, according to equation (5) where $v_i^\sigma(a_i,x)$ are computed using arguments P in place of $P_i^\sigma(a_i|x)$. The MPE equilibrium constitutes a fixed point of Λ, i.e. $P^\star = \Lambda(P^\star)$, existence of which is guaranteed by the Brouwer theorem. However, unlike the Bellman operator above, the *best response probability operator* Λ is not a contraction and may have multiple fixed points. Effectively, the equilibrium choice probabilities $P^\star$ solve N Bellman equations simultaneously.

For completeness, we present an alternative representation of the best response mapping that facilitates the definition of the directional dynamic games in the next section. Applying the law of iterated expectations to the equation (2) we can rewrite it as

$$V_i^\sigma(x) = \sum_{a_i \in A} P_i^\sigma(a_i|x) \left[\pi_i^\sigma(a_i,x) + e_i^\sigma(a_i,x) + \beta \sum_{x' \in X} V_i^\sigma(x') f_i^\sigma(x'|a_i,x) \right], \tag{6}$$

where the term $e_i^\sigma(a_i,x) \equiv E(\varepsilon_i[a_i]|x,\sigma(x,\varepsilon_i) = a_i)$ is the expectation of the private shock ε_i conditional on action a_i being assigned by strategy σ_i in state x. Hotz and Miller (1993) show that this term is a function only of the choice probabilities $P_i^\star(a_i|x)$ and the private shock distribution g_i. Under (EV), it takes the form $e_i^\sigma(a_i,x) = \eta\gamma - \eta\log(P_i^\sigma(a_i|x))$.

Recognizing that for every $i \in \{1,\dots,N\}$ the equation (6) represents a linear system of equations in the unknowns $V_i^\sigma(x)$, we can rewrite and solve it in matrix form. Let V_i^σ stack all values of $V_i^\sigma(x)$ over the state space to form a $|X| \times 1$ column vector, and define $\pi_i^\sigma(a_i)$, $e_i^\sigma(a_i)$ and $P_i^\sigma(a_i)$ similarly. The solution of the linear system (6) is given by

$$V_i^\sigma = (I - \beta F^\sigma)^{-1} \sum_{a_i \in A} P_i^\sigma(a_i) * [\pi_i^\sigma(a_i) + e_i^\sigma(a_i)], \tag{7}$$

where $*$ denotes the element-wise Hadamard product, I is the identity matrix and F^σ is defined by the following.

Definition 2 (Transition probablity matrices)**.** *The conditional transition matrix $F(a)$ is a $|X| \times |X|$ matrix with a typical element $f(x'|a,x)$ equal to the transition probability from common knowledge state x to state x' given the action profile a. The unconditional transition matrix F^σ is a $|X| \times |X|$ matrix with a typical element $F^\sigma_{x,x'} = \sum_{a \in A^N} \left(\prod_{i=1}^N P_i^\sigma(a_i|x)\right) f(x'|a,x)$ equal to the transition probability from common knowledge state x to state x' averaged over all possible action profiles a with the corresponding joint choice probabilities.*

The unconditional transition matrix F^σ can also be computed as a weighted sum of the conditional transition matrices $F(a)$ with the similar weights, which equal to the joint probability of action profile a being realized under strategy profile σ. Under assumptions (CI) and (IPV) this probability is given by $\prod_{i=1}^N P_i^\sigma(a_i|x)$. Consequently, F^σ is the one-step transition matrix of the Markov chain formed by the common knowledge state $\{x_t\}$ under the strategy profile σ. We suppress the superscript and write F whenever the dependence on σ is not essential.

The right hand side of equation (7) defines a *policy valuation* operator $\Gamma\colon P \mapsto V^\sigma$, which maps the space of choice probabilities into the space of value functions implementing the Hotz-Miller inversion. Define the $\Psi\colon P \mapsto P$ operator by the composition $P \mapsto V^\sigma \mapsto v^\sigma \mapsto P$ which maps choice probabilities for all (a,x) for all players $i \in \{1,\dots,N\}$ to the corresponding value functions using Γ operator, then choice specific value functions using equation (3), and finally back to the choice probabilities using equation (5). (AM2007, Representation Lemma) establishes that the MPE equilibria of the game can be equivalently characterized as fixed points of both Λ and Ψ operator, i.e. $P^\star = \Psi(P^\star) = \Lambda(P^\star)$, with computational implementation of the Ψ operator being less costly.

2.2 Directionality in dynamic games

IRS2016 introduced *directional dynamic games* (DDG) as a subclass of discrete stochastic games that possess an additional structure in the state space. They considered the case when the state space X can be decomposed as $X = D \times X'$, and formalized the directionality of the game in terms of the probabilities of eventually reaching the points of D from one another conditional on the strategy profile σ. We adopt a closely related alternative definition cast in terms of the graph of positive *one-step* transitions between the points of D, which we then characterize through the block structure of the transition probability matrices F^{σ} and $F(a)$.

Given the decomposition $X = D \times X'$, write $x = (d, x')$ for a generic state, and for $d \in D$ let $B_d = \{d\} \times X'$ denote the *substage* associated with d, so that the substages $\{B_d : d \in D\}$ partition the state space into $|D|$ cells of the common size $|X'|$. Let Σ denote the set of feasible Markovian strategy profiles σ introduced in the previous section, and recall from Definition 2 that F^{σ} is the one-step transition matrix of the common knowledge state under the profile σ. For $\sigma \in \Sigma$, the *transition graph* $\mathscr{D}(\sigma)$ is then defined as the directed graph whose vertices are the elements of D and which contains an edge $d \to d'$ between two distinct vertices $d \neq d'$ if and only if $F^{\sigma}_{x,x'} > 0$ for some $x \in B_d$ and $x' \in B_{d'}$; self-loops $d \to d$ are ignored by construction.

Definition 3 (Directional dynamic game)**.** *A discrete dynamic game of incomplete information with the state space decomposed as $X = D \times X'$ is* directional *with respect to this decomposition if (1) the transition graph $\mathscr{D}(\sigma)$ is acyclic for every $\sigma \in \Sigma$, and (2) no two graphs $\mathscr{D}(\sigma)$ and $\mathscr{D}(\sigma')$, $\sigma, \sigma' \in \Sigma$, contain edges in opposite orientation between any two vertices.*[2]

Call a strategy profile *completely mixed* if $P_i^{\sigma}(a_i|x) > 0$ for all i, a_i and x. Completely mixed profiles exist under the maintained assumptions,[3] and by Definition 2 the set of non-zero elements of $F^{\bar{\sigma}}$ for any completely mixed $\bar{\sigma}$ is the largest among all $\sigma \in \Sigma$, namely the union of the sets of non-zero elements of the conditional transition matrices $F(a)$, $a \in A^N$: all terms in the defining sum are non-negative, and at a completely mixed profile every action profile receives a strictly positive weight.

The *maximal transition graph* $\bar{\mathscr{D}}$ is the directed graph whose vertices are the elements of D and which contains an edge $d \to d'$ between two distinct vertices $d \neq d'$ if and only if $f(x'|a,x) > 0$ for some $a \in A^N$, $x \in B_d$ and $x' \in B_{d'}$. It follows from the support maximality above that $\bar{\mathscr{D}} = \mathscr{D}(\bar{\sigma})$ for

[2] The one-step conditions (1) and (2) of Definition 3 imply the original hitting probability based definition of IRS2016 but are strictly stronger, see Lemma A.1 and the discussion in Appendix A.1.

[3] The marginal distribution of the first component $\varepsilon_i[0]$ of the vector of private shocks is absolutely continuous, and therefore atomless, so there exist quantiles $-\infty = \bar{q}_0 \leq \bar{q}_1 \leq \cdots \leq \bar{q}_J \leq \bar{q}_{J+1} = +\infty$ with $Pr\{\bar{q}_a < \varepsilon_i[0] \leq \bar{q}_{a+1}\} = 1/(J+1)$ for every $a \in A$. The profile $\bar{\sigma}$ with decision rules $\bar{\sigma}_i(x, \varepsilon_i) = a$ whenever $\bar{q}_a < \varepsilon_i[0] \leq \bar{q}_{a+1}$ is feasible and satisfies $P_i^{\bar{\sigma}}(a_i|x) = 1/(J+1) > 0$ for all a_i and x.

every completely mixed profile $\bar{\sigma}$, and that every transition graph $\mathscr{D}(\sigma)$, $\sigma \in \Sigma$, is a subgraph of $\bar{\mathscr{D}}$; in particular, in a directional game $\bar{\mathscr{D}} = \mathscr{D}(\bar{\sigma})$ is acyclic by condition (1) of Definition 3, justifying the terminology.[4]

Finally, we say that F^{σ} is *upper block-triangular with respect to a total order* $\trianglelefteq$ on D if

$$F^{\sigma}_{x,x'} = 0 \quad \text{whenever} \quad x \in B_d,\ x' \in B_{d'} \ \text{ and } \ d' \lhd d, \tag{8}$$

where $d' \lhd d$ denotes $d' \trianglelefteq d$, $d' \neq d$. In words, listing the states of X substage by substage in the order given by $\trianglelefteq$ (with arbitrary ordering within each substage) is a simultaneous permutation of the rows and columns of F^{σ} after which all non-zero elements are contained in the $|D|$ diagonal blocks of the common size $|X'|$ and the blocks above them; the diagonal blocks themselves are unrestricted, so arbitrary transitions *within* a substage, including self-loops, are compatible with (8).

Proposition 2.1 (Directional dynamic games of incomplete information)**.** *A discrete dynamic game of incomplete information of Section 2.1 with the state space decomposed as $X = D \times X'$ is directional with respect to this decomposition (Definition 3) if and only if there exists a total order $\trianglelefteq$ on D such that the unconditional transition probability matrix F^{σ} is upper block-triangular (8) for every feasible strategy profile $\sigma \in \Sigma$, or, equivalently, for a single completely mixed profile $\bar{\sigma}$.*

Proof. See Appendix A.1. □

Corollary 2.2. *The game is directional with respect to the decomposition $X = D \times X'$ if and only if all conditional transition probability matrices $F(a)$, $a \in A^N$, are upper block-triangular (8) with respect to a common total order on D.*

Proof. Since the set of non-zero elements of $F^{\bar{\sigma}}$ at a completely mixed profile $\bar{\sigma}$ is the union of the sets of non-zero elements of $\{F(a) : a \in A^N\}$, as established above, condition (8) holds for every $F(a)$ with respect to a given total order if and only if it holds for $F^{\bar{\sigma}}$. The claim then follows from Proposition 2.1 applied at $\bar{\sigma}$. □

Note that by Corollary 2.2 directionality can be verified in a single pass over the model primitives: what needs to be examined is only the common zero pattern of the conditional matrices $F(a)$ — equivalently, the support of $\sum_{a \in A^N} F(a)$ — which is a property of the transition function f alone and does not depend on the strategies.[5] Lemma A.1 in the Appendix also derives directly from conditions

[4] Note also that condition (2) of Definition 3 is implied by condition (1): a pair of edges in opposite orientation in any two graphs $\mathscr{D}(\sigma)$ and $\mathscr{D}(\sigma')$ would form a directed cycle of length two in $\bar{\mathscr{D}}$, which is acyclic as just noted.

[5] Note also that the total order $\trianglelefteq$ in Proposition 2.1 is in general not unique: any topological extension of the strategy-independent partial order on D constructed in Lemma A.1 satisfies (8), and the choice among the extensions affects only the sequence in which the solution algorithm processes non-comparable substages, not the set of equilibria it computes.

(1) and (2) of Definition 3 the two properties that underlie the theory of IRS2016: the existence of a strategy-independent strict partial order on D that refines every strategy-specific order, and the monotonicity of play with respect to it under every feasible strategy profile. The stage decomposition, state recursion and recursive lexicographical search results of IRS2016, which build on these two properties, therefore apply to the directional games characterized by Proposition 2.1.

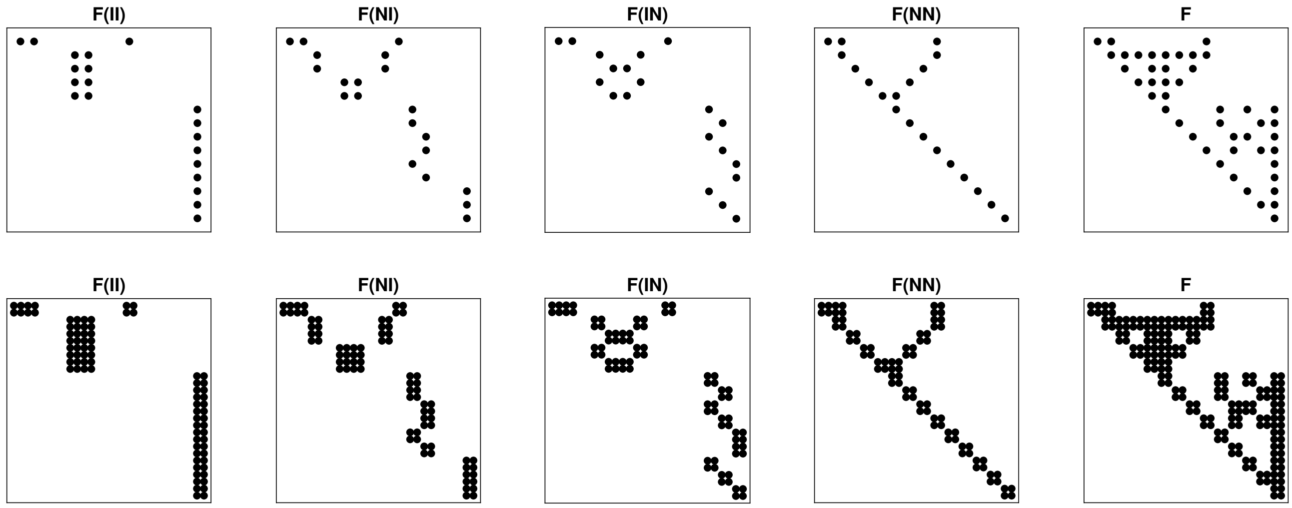
2.3 Example: the leapfrogging game

The empirical literature on dynamic games has multiple examples of DDGs, including platform competition (Dube, Hitsch and Chintagunta, 2010), patent races (Judd, Schmedders and Yeltekin, 2012), and tennis games (Anderson, Rosen, Rust and Wong, 2025) among others. Throughout this paper we use the dynamic model of Bertrand competition with cost-reducing investments studied in IRS2018 as a running example of a directional dynamic game (DDG). The typical equilibrium in this model involves the two asymmetric duopolists leapfrogging each other in the marginal cost of production, hence the model's shorthand name. Depending on the specification and parameter values, this relatively simple game has from a unique equilibrium (IRS2018, Theorem 4) up to hundreds of millions of equilibria in computed examples, making it an ideal testbed for studying the effects of multiplicity on the performance of estimators of dynamic games; we use it as the data generating process in the Monte Carlo experiments of Section 4.

In the context of the discrete dynamic games of incomplete information, consider the game between $N = 2$ Bertrand competitors who interact over time by *simultaneously* deciding whether or not to invest in the latest (state of the art) production technology c at a cost $K(c)$, that is $A = \{I, N\}$ corresponding to the two options. We parameterize the investment cost as $K(c) = k_1/(1 + ck_2)$, $k_1, k_2 > 0$, such that the cost of investment increases as the state-of-the-art cost of production c falls. If either of the firms invests, their current technology with marginal cost of production c_i is replaced by a better technology with lower cost c after one period lag. The common knowledge state of the game is given by $x = (c_1, c_2, c)$ where the state-of-the-art cost c evolves exogenously according to a Markov transition process $f_c(c'|c)$ where $f_c(c'|c) = 0$ for all $c' > c$ reflecting the non-reversibility of technological progress. Assuming myopic demand and normalizing the market size to one, the instantaneous payoffs follow the outcome of the static Bertrand competition with the indirect profit function given by

$$\Pi_i(a,x) = (c_j - c_i)\mathbb{1}\{c_i < c_j\} - K(c)\mathbb{1}\{a_i = I\} = \pi_i(a_i,x),\ i, j \in \{1,2\}, i \neq j. \tag{9}$$

Figure 1: Conditional and unconditional transition probability matrices in the leapfrogging game.



Notes: The top row of panels presents the non-zero elements of the conditional and unconditional transition probability matrices $F(a)$ and F in the simultaneous move version of the leapfrogging game with $n = 3$, and the bottom row presents the same matrices in the alternating move version of the same game. The action space is given by $\{I, N\}$ for each of the two players, resulting in four possible combinations of actions. Number of points in the state space is 14 and 28 for the two versions of the game, respectively, where in the alternating move game the additional state variable $m \in \{1, 2\}$ is not directional, resulting in 2-by-2 blocks in the transition probability matrices.

The tie-breaking rule for the case $c_1 = c_2$ does not affect the payoffs, which are zero for both firms. The last equality is due to the time-to-build assumption ensuring that current actions are not affecting the current payoffs, see equation (1).

For generality we also consider the case of differentiated products where consumers choose the lower price subject to an idiosyncratic EV1 shock with scale λ, such that the probability of choosing product i is given by the logit formula $q_i(p_1^\lambda, p_2^\lambda, \lambda) = \exp(-p_i^\lambda/\lambda) / \big(\exp(-p_1^\lambda/\lambda) + \exp(-p_2^\lambda/\lambda)\big)$. In this case, the prices $(p_1^\lambda, p_2^\lambda)$ charged by the two firms constitute Bayesian-Nash equilibrium in a static pricing game, and are given as a solution to the following system of equations

$$
\begin{aligned}
p_1^\lambda &= \arg\max\nolimits_p q_1(p, p_2^\lambda, \lambda)(p - c_1), \\
p_2^\lambda &= \arg\max\nolimits_p q_2(p_1^\lambda, p, \lambda)(p - c_2),
\end{aligned}
\tag{10}
$$

in which case the indirect profit functions are given by $\Pi_i^\lambda(a, x) = q_i(p_1^\lambda, p_2^\lambda, \lambda)(p_i^\lambda - c_i) - K(c)\mathbb{1}\{a_i = I\}$. Even though technically the strategies in this case also include pricing decisions, we treat the profit functions implied by the static Bayesian-Nash equilibrium as indirect payoffs and continue focusing on the investment decisions $a_i \in A = \{I, N\}$ as the only strategic actions. It is straightforward to show that as $\lambda \to 0$, the differentiated product case converges to pure Bertrand competition with $p_1^\lambda \to \max(c_1, c_2)$, $p_2^\lambda \to \max(c_1, c_2)$, and $\Pi_i^\lambda(a, x) \to \Pi_i(a, x)$ given in equation (9).

The state space X of the leapfrogging game is discretized with n points in each of three dimensions, and is given by $X = \{(c_1, c_2, c) \in \mathbb{R}^3 : c_1 \geqslant c, c_2 \geqslant c\}$, incorporating the natural technological constraints. The cardinality of the state space is consequently equal $|X| = n(n+1)(2n+1)/6$. Whereas c evolves as an exogenous Markov process, transitions of (c_1, c_2) are governed directly by the investment decisions of the two firms: for both i the decision $a_i = N$ causes $c_i' = c_i$, and the decision $a_i = I$ causes $c_i' = c \leqslant c_i$. The top row of Figure 1 shows the sparsity structure of the conditional and unconditional transition probabilities $F(a)$ and F per Definition 2 for the leapfrogging game with simultaneous moves, $n = 3$. The total order $\trianglelefteq$ in (8) on the state space used is given by the following rules: the state point (c_1, c_2, c) precedes point (c_1', c_2', c') if (a) $c > c'$; otherwise if (b) $c_1 > c$ and $c_2 > c$, otherwise if (c) $c_1 = c$ or $c_2 = c$ but $c_1 \neq c_2$. The four conditional transition probability matrices $F(a)$ for $a \in \{(I,I), (N,I), (I,N), (N,N)\}$ and the unconditional F clearly have upper triangular form (block-triangular with 1×1 blocks). By Corollary 2.2, this common triangular form verifies that the simultaneous move leapfrogging game satisfies the one-step directionality conditions of Definition 3 with the fully directional state, $D = X$ and singleton X': no transition can increase any of the cost variables, so the maximal transition graph $\bar{\mathscr{D}}$ is acyclic.

We also consider the *alternating move* version of the leapfrogging game, where the right to make the investment decision is given to one firm at a time by an additional state variable $m \in \{1, 2\}$ that evolves as an exogenous Markov chain with transition probabilities $f(m'|m)$, allowing for both deterministic alternation and random moves. The directional component (c_1, c_2, c) is unchanged, while m constitutes the non-directional component of the state space, $X' = \{1, 2\}$: with the state points indexed such that the m-dimension is the last, the transition probability matrices acquire 2×2 diagonal blocks corresponding to each element of the simultaneous move matrices, as shown in the bottom row of Figure 1, and the game remains directional by Proposition 2.1.

More details on the setup of the leapfrogging game can be found in IRS2016, with the theoretical properties of the model investigated in IRS2018. The Online Appendix contains the full specification of both simultaneous and alternating move versions of the leapfrogging game along with the description of the solution algorithm for the stage games discussed in the next section.

3 Full solution maximum likelihood estimator

In this section we define the full solution maximum likelihood estimator for the directional dynamic games and establish its statistical properties. The estimator is based on repeated solutions of the dynamic game using the recursive lexicographic search (RLS) algorithm of IRS2016, therefore in line with nested fixed point estimator of Rust (1987) we term it *nested RLS* estimator (NRLS). We first present the general framework for the estimator and its asymptotic properties, and then describe the numerical implementation of the estimator in more detail. The main idea of the tractable numerical implementation is to rely on the branch-and-bound (BnB) algorithm (Land and Doig, 1960) in the nested maximization loop to avoid computing those equilibria which are unlikely in the given sample. The RLS algorithm is reinterpreted as tree traversal algorithm to match the BnB approach.

3.1 General framework

Consider M markets, each with N players indexed by $i \in \{1,\dots,N\}$, observed over T time periods. In market $m \in \{1,\dots,M\}$ at time $t \geq 1$, let $x_{mt} \in X$ denote the common knowledge state and $a_{mt} = (a_{1mt},\dots,a_{Nmt})$ the action profile of the players, where the individual actions $a_{imt} \in A = \{0,\dots,J\}$. For now we treat the dynamics of x_{mt} as known (pre–estimated).

To facilitate the estimation problem, we make the dependence of the model on the structural parameters explicit. Let the model primitives, in particular the deterministic payoffs $\Pi_i(a,x) = \Pi_i(a,x;\theta)$ and the transition probabilities $f(x'|a,x) = f(x'|a,x;\theta)$, be known functions of a finite-dimensional vector of structural parameters $\theta \in \Theta$. All objects derived in Section 2.1, including the choice specific value functions, the choice probabilities and the operators Λ and Ψ, inherit the dependence on θ. For a given θ, let

$$\mathscr{E}(\theta) = \{P : P = \Psi(P;\theta)\} \tag{11}$$

denote the set of all MPE of the game, which is non-empty (AM2007), but in general is not a singleton since neither Λ nor Ψ is a contraction. For an equilibrium $e \in \mathscr{E}(\theta)$ we denote by $P_i(a_i|x;e,\theta)$, $i \in \{1,\dots,N\}$, $a_i \in A$, the associated equilibrium choice probabilities: these are the objects from which the likelihood function of the observed actions and states is constructed.

Importantly, the set of possible equilibria (11) is parameter–dependent. Let $\mathscr{E} = \bigcup_{\theta\in\Theta}\mathscr{E}(\theta) = \{e_1,\dots,e_L\}$ be the set of all L equilibria that exist for at least one value of $\theta \in \Theta$, so that $\mathscr{E}(\theta) \subseteq \mathscr{E}$ is the subset that exists for a particular value of θ. Finally, let $\Theta(e) = \{\theta \in \Theta : e \in \mathscr{E}(\theta)\} \subseteq \Theta$ be the parameter subset for which equilibrium $e \in \mathscr{E}$ exists.

Given a random sample of markets, each of which is observed over T time periods, a_{mt}, x_{mt}, $1 \leq m \leq M$ and $1 \leq t \leq T$, we can then express the log-likelihood conditional on θ being the data-generating structural parameter value and $e_{1:M} = (e_m)_{m=1}^{M} \in \mathscr{E}^M$ being the profile of equilibria played across the M markets as

$$\mathscr{L}_{MT}(\theta, e_{1:M}) = \frac{1}{MT}\sum_{m=1}^{M}\sum_{t=1}^{T}\ell(z_{mt};\theta,e_m),\ \ell(z_{mt};\theta,e) = \sum_{i=1}^{N}\log P_i(a_{imt}|x_{mt};e,\theta), \tag{12}$$

where $z_{mt} = (a_{mt}, x_{mt})$, and $P_i(a_i|x;e,\theta)$ are the equilibrium choice probabilities defined above. We can here think of the particular equilibria played across the different markets, $e_{1:M}$, as unknown market specific fixed effects. Note that $\mathscr{L}_{MT}(\theta, e_{1:M})$ is not well-defined for $\theta \notin \bigcap_{m=1}^{M}\Theta(e_m)$. We therefore define the parameter set as $\mathscr{B}_M = \left\{(\theta, e_{1:M}) \in \Theta \times \mathscr{E}^M : \theta \in \bigcap_{m=1}^{M}\Theta(e_m)\right\}$ and the corresponding maximum–likelihood estimator as

$$(\hat{\theta}, \hat{e}_{1:M}) = \arg\max_{(\theta, e_{1:M})\in\mathscr{B}_M}\mathscr{L}_{MT}(\theta, e_{1:M}). \tag{13}$$

Finally, note that if the state dynamics contain unknown parameters, say, θ_x, to be estimated then

$$\mathscr{L}_{MT}^{(x)}(\theta_x) = \frac{1}{MT}\sum_{m=1}^{M}\sum_{t=1}^{T}\log f(x_{mt}|x_{m,t-1}, a_{m,t-1};\theta_x) \tag{14}$$

needs to be added to $\mathscr{L}_{MT}(\theta, e_{1:M})$ in equation (12).

Same equilibrium across all markets

Consider first the scenario where the same equilibrium is played in all markets. Under this scenario the data requirements are quite weak. In particular, we can let T stay fixed and only let $M \to \infty$. As a consequence, we do not need to make any assumptions about the time series properties of z_{mt}; in particular, it is allowed to be non–stationary and so the theory covers directional games.

Let a single equilibrium $e_0 \in \mathscr{E}$ be played across all markets. Suppose furthermore that the econometrician knows this and so computes the MLE estimator as

$$(\hat{\theta}, \hat{e}) = \arg\max_{(\theta,e)\in\mathscr{B}}\mathscr{L}_{MT}(\theta, e), \tag{15}$$

where $\mathscr{L}_{MT}(\theta, e) = \mathscr{L}_{MT}(\theta, e, ..., e)$, and $\mathscr{B} = \{(\theta, e) \in \Theta \times \mathscr{E} : \theta \in \Theta(e)\}$. Let

$$\mathscr{L}_T(\theta, e) = E[\mathscr{L}_{MT}(\theta, e)] = \frac{1}{T}\sum_{t=1}^{T}E[\ell(z_{mt};\theta,e)]$$

denote the asymptotic limit of $\mathscr{L}_{MT}(\theta, e)$ as $M \to \infty$. We impose the following assumption on the model:

Assumption B. *(i) Θ is compact and $\mathscr{E}$ is finite with $\theta_0 \in \Theta(e_0)$ and $e_0 \in \mathscr{E}$; (ii) $\mathscr{L}_T(\theta, e) < \mathscr{L}_T(\theta_0, e_0)$ for any $\theta \in \Theta(e)$ and $e \in \mathscr{E}$ with $(\theta, e) \neq (\theta_0, e_0)$; (iii) $\theta \mapsto \ell(z_{mt}; \theta, e)$ is almost surely continuous on $\Theta(e)$ for all $e \in \mathscr{E}$; (iv) $\|\ell(z_{mt}; \theta, e)\| \leq b(z_{mt})$ for all $\theta \in \Theta(e)$ and $e \in \mathscr{E}$, where $\sum_{t=1}^{T} E[b(z_{mt})] < \infty$.*

Assumption B is fairly standard for showing consistency of M–estimators, c.f. Section 2 of Newey and McFadden (1994). Parts (i)-(ii) ensures that the data–generating parameter value θ_0 and equilibrium e_0 are well-separated from the other potential ones of the model and so are identified. Finiteness of $\mathscr{E}$ is guaranteed in finite dynamic stochastic games by Haller and Lagunoff (2000) and Doraszelski and Escobar (2010) for generic payoffs. Parts (i) and (iii)-(iv) imply that $\mathscr{L}_{MT}(\theta, e)$ converges uniformly in (θ, e) towards $\mathscr{L}_T(\theta, e)$ in probability.

Theorem 3.1 (Consistency)**.** *Under Assumption B, $\hat{\theta} \to^P \theta_0$ and $\Pr(\hat{e} = e_0) \to 1$, as $M \to \infty$.*

Proof. See Appendix A.2. □

Note here that $\hat{e} = e_0$ w.p.a.1. Thus, we can treat e_0 as known when showing asymptotic normality of $\hat{\theta}$. That is, $\hat{\theta}$ is first–order equivalent to the oracle estimator where the true equilibrium is known to us. Let $s(z_{mt}; \theta, e) = \frac{\partial \ell(z_{mt}; \theta, e)}{\partial \theta}$ and $h(z_{mt}; \theta, e) = \frac{\partial^2 \ell(z_{mt}; \theta, e)}{\partial \theta \partial \theta'}$ denote the score and Hessian of the log-likelihood function, respectively, and let $H_T(\theta, e) = \frac{1}{T} \sum_{t=1}^{T} E[h(z_{mt}; \theta, e)]$. For the asymptotic normality of $\hat{\theta}$ we impose the following additional assumption:

Assumption C. *(i) $\theta_0 \in \mathrm{int}\Theta(e_0)$; (ii) $\theta \mapsto \ell(z_{mt}; \theta, e_0)$ is almost surely twice continuously differentiable in a neighborhood of θ_0; (iii) $\sum_{t=1}^{T} E[s(z_{mt}; \theta_0, e)] = 0$ and $\sum_{t=1}^{T} E\left[\|s(z_{mt}; \theta_0, e_0)\|^2\right] < \infty$; $\|h(z_{mt}; \theta, e_0)\| \leq b(z_{mt})$ where $\sum_{t=1}^{T} E[b(z_{mt})] < \infty$ for all θ in a neighborhood of θ_0 in $\Theta(e_0)$, and $H_T(\theta_0, e_0)$ has full rank.*

Theorem 3.2 (Asymptotic normality)**.** *Let $\hat{\theta}_0 = \arg\max_{\theta \in \Theta(e_0)} \mathscr{L}_{MT}(\theta, e_0)$ be the oracle estimator. Under Assumption B, $\Pr\left(\hat{\theta} = \hat{\theta}_0\right) \to 1$. If in addition Assumption C also holds,*

$$\sqrt{M}(\hat{\theta} - \theta_0) = \sqrt{M}(\hat{\theta}_0 - \theta_0) + o_P(1) \to^d N\left(0, H_{T,0}^{-1} \Omega_{T,0} H_{T,0}^{-1}\right), \tag{16}$$

where $H_{T,0} = H_T(\theta_0, e_0)$ and $\Omega_{T,0} = \sum_{t_1, t_2 = 1}^{T} E\left[s(z_{m,t_1}; \theta_0, e_0) s(z_{m,t_2}; \theta_0, e_0)'\right] / T^2$. If the model is correctly specified then $\Omega_{T,0} = H_{T,0}$.

Proof. See Appendix A.3. □

Above theorem assumes that $\theta_0 \in \mathrm{int}\Theta(e_0)$, c.f. Assumption C(i). If instead θ_0 is situated on the boundary of $\Theta(e_0)$ then the asymptotic distribution of the MLE becomes non–standard; see, e.g., Andrews (1999). The distribution depends on the curvature of the boundary, and so in order to develop such a theory, additional assumptions about the form of $\Theta(e_0)$ have to be imposed. If, for example, θ is a scalar and $\Theta(e_0) = [\theta_-, \theta_+]$ with $\theta_0 = \theta_+$ situated on the boundary then $\sqrt{M}(\hat{\theta} - \theta_0) \to^d \min\{0, W\}$, where $W \sim N\left(0, H_{T,0}^{-1}\Omega_{T,0}H_{T,0}^{-1}\right)$.[6]

The above analysis relies on that e_0 is well–separated from the other equilibria of the model in the population. This in turn implies that in large samples, we can treat e_0 as known and do not have to take into account any uncertainty about the true equilibrium. In small samples, however, the log-likelihood evaluated at two different equilibria may be close to indistinguishable. The following proposition, proved in Appendix A.4, characterizes the resulting uncertainty about the selected equilibrium. For each $e \in \mathscr{E}$ let $\hat{\theta}(e) = \arg\max_{\theta \in \Theta(e)} \mathscr{L}_{MT}(\theta, e)$ and $\theta_0(e) = \arg\max_{\theta \in \Theta(e)} \mathscr{L}_T(\theta, e)$ denote the equilibrium–specific estimator and pseudo–true value, so that the estimator (15) satisfies $\hat{e} = \arg\max_{e \in \mathscr{E}} \mathscr{L}_{MT}(\hat{\theta}(e), e)$.

Assumption D. *For every $e \in \mathscr{E}$: (i) $\theta \mapsto \ell(z_{mt}; \theta, e)$ is almost surely continuous on the closure of $\Theta(e)$, over which $\theta_0(e)$ is the unique maximizer of $\mathscr{L}_T(\theta, e)$, and $\theta_0(e) \in \operatorname{int}\Theta(e)$; (ii) Assumption C(ii)–(iii) holds with $(\theta_0(e), e)$ in place of (θ_0, e_0); (iii) $\sum_{t=1}^T E\left[b(z_{mt})^2\right] < \infty$ for the dominating function b of Assumption B(iv).*

Proposition 3.3 (Equilibrium selection as a multinomial probit)**.** *Suppose Assumptions B and D hold. Then: (i) jointly across $e \in \mathscr{E}$,*

$$Z_M(e) = \sqrt{M}\left\{\mathscr{L}_{MT}(\hat{\theta}(e), e) - \mathscr{L}_T(\theta_0(e), e)\right\} \to^d Z \sim N(0, \Sigma_T), \tag{17}$$

where $\Sigma_T(e, e') = \operatorname{Cov}\left(\frac{1}{T}\sum_{t=1}^T \ell(z_{mt}; \theta_0(e), e), \frac{1}{T}\sum_{t=1}^T \ell(z_{mt}; \theta_0(e'), e')\right)$;
(ii) if in addition $\operatorname{Var}(Z(e) - Z(e')) > 0$ for every pair $e \neq e'$ with $\mathscr{L}_T(\theta_0(e), e) = \mathscr{L}_T(\theta_0(e'), e')$, then for every $e \in \mathscr{E}$, as $M \to \infty$,

$$\Pr(\hat{e} = e) - \Pr(\hat{e}_M^* = e) \to 0, \ \ \hat{e}_M^* = \arg\max_{e \in \mathscr{E}} \left\{\sqrt{M}\,\mathscr{L}_T(\theta_0(e), e) + Z(e)\right\}. \tag{18}$$

That is, in large samples the selected equilibrium behaves as the choice in a multinomial probit model in which equilibrium e carries the deterministic "utility" $\sqrt{M}\mathscr{L}_T(\theta_0(e), e)$ and the normally distributed "taste shock" $Z(e)$, so the sampling distribution of $\hat{e}$ can be computed with standard simulation methods for the multinomial probit, with the utilities and Σ_T replaced by their sample counterparts: the deterministic component $\mathscr{L}_T(\theta_0(e), e)$ is consistently estimated by $\mathscr{L}_{MT}(\hat{\theta}(e), e)$, and Σ_T by the sample covariance matrix of the market–level vectors $\left(\frac{1}{T}\sum_{t=1}^T \ell(z_{mt}; \hat{\theta}(e), e)\right)_{e \in \mathscr{E}}$, $m \in$

[6] See Section 4.3 for this detail showing in Monte Carlo simulations.

$\{1,\ldots,M\}$. Given these estimates, the choice probabilities $\Pr(\hat{e}^*_M = e)$ can be computed using existing simulation–based methods for the multinomial probit; see, e.g., Börsch-Supan and Hajivassiliou (1993). As $M \to \infty$ the probit probabilities collapse on the well–separated e_0, recovering Theorem 3.1; at moderate M they quantify precisely the near–indistinguishability of equilibria that makes small–sample equilibrium selection delicate.

Multiple equilibria played in the data

Consider now the case where multiple equilibria are played across the different markets. We here discuss different estimators that can be used in this setting. As before θ_0 denote the true value of the structural parameters while $e^0_{1:M} = (e_{0m})_{m=1}^M$ denotes the equilibria played across the M markets.

The most robust estimation method is to treat the equilibria played in the different markets as fixed effects leading to the following estimator,

$$(\hat{\theta}, \hat{e}_{1:M}) = \arg \max_{(\theta, e_{1:M}) \in \mathscr{B}_M} \mathscr{L}_{MT}(\theta, e_{1:M}). \tag{19}$$

This approach is readily implemented by the NRLS algorithm described in Section 3, but is computationally demanding since each equilibrium in $\mathscr{E}(\theta)$ has to be evaluated for each market. As it will be shown the computational burden is reduced in larger samples which is this case is effectively given by $T \ll MT$. But given that the RLS algorithm is able to produce the set of all equilibria $\mathscr{E}(\theta)$ for a given θ the estimator (19) is still feasible for a moderate number of markets and players.[7]

If we allow different *known* groups of markets to play different equilibria, let $\{\mathscr{M}_g\}_{g=1}^G$ be a known partition of the M markets into G groups, let $M_g = |\mathscr{M}_g|$, and suppose all markets in group g play the same equilibrium $e_g \in \mathscr{E}(\theta)$. Equilibrium play remains constant within each market over time. Conditional on the partition, the grouped full-solution likelihood estimator is

$$(\hat{\theta}, \hat{e}_{1:G}) = \arg \max_{(\theta, e_{1:G}) \in \mathscr{B}_G} \frac{1}{MT} \sum_{g=1}^G \sum_{m \in \mathscr{M}_g} \sum_{t=1}^T \ell(z_{mt}; \theta, e_g), \tag{20}$$

where $\mathscr{B}_G = \left\{(\theta, e_{1:G}) \in \Theta \times \mathscr{E}^G : \theta \in \bigcap_{g=1}^G \Theta(e_g)\right\}$ and $\ell(z_{mt}; \theta, e)$ is defined in (12).

This is a grouped restriction of the multiple-equilibrium likelihood in (19): instead of estimating a separate e_m for every market, it imposes $e_m = e_g$ for all $m \in \mathscr{M}_g$. With fixed T, the equilibrium

[7] Hahn and Moon (2010) provide an asymptotic theory for this estimator in the case where z_{mt} is stationary and mixing. This property is essential for their analysis since it implies that the estimated equilibrium in a given market is consistent as $T \to \infty$ under regularity conditions similar to the ones found in Assumption B together with stationarity. However, their stationarity assumption is violated in our directional games since x_{mt} has an absorbing state which it reaches in finite time with probability 1. This in turn implies that their fixed effects estimator of the equilibrium played in market m cannot be consistent in our setting.

played by one directional market cannot generally be classified consistently from its finite history. The grouped design instead pools M_g markets behind each e_g; with G fixed and M_g growing, the group-specific likelihood contains increasing information about its equilibrium assignment. In an application, groups could be defined from observed market characteristics or, as a separate first step, estimated by clustering.

3.2 Nested RLS estimator (NRLS)

The MLE estimator developed in the previous section is highly computationally demanding, as it requires the computation of the choice probabilities $P_i(a_{imt}|x_{mt};e_m,\theta)$ for each player and action on the set $\mathscr{E}(\theta)$ of all MPE in the game for each trial value of the structural parameters θ.[8] We therefore focus on the computational aspects of the proposed estimator in the next three sections, and show how the computational burden can be substantially reduced, making the estimator computationally feasible.

First, again abstracting from the likelihood of state dynamics in (14), we represent the maximization problems (13) (under the assumption of a single equilibrium played in all markets) and (19) (under the assumption that different equilibria are played in different markets) in the following nested form

$$\hat{\theta} = \arg\max_{\theta\in\Theta} \mathscr{L}_{MT}(\theta),\ \mathscr{L}_{MT}(\theta) = \begin{cases} \max_{e\in\mathscr{E}(\theta)} \frac{1}{M}\sum_{m=1}^{M} \mathscr{L}_T(m,\theta,e), & \text{[single]} \\ \max_{e_{1:M}\in\mathscr{E}(\theta)^M} \frac{1}{M}\sum_{m=1}^{M} \mathscr{L}_T(m,\theta,e_m), & \text{[multiple]} \\ \max_{e_{1:G}\in\mathscr{E}(\theta)^G} \frac{1}{M}\sum_{g=1}^{G}\sum_{m\in\mathscr{M}_g} \mathscr{L}_T(m,\theta,e_g), & \text{[grouped]} \end{cases} \quad (21)$$

with the estimates $\hat{e}$, $\hat{e}_{1:M}$ or $\hat{e}_{1:G}$ of the played equilibria given by the arg max of the inner problem. The market-specific likelihood function $\mathscr{L}_T(m,\theta,e)$ is defined in (22) below.

Thus, the proposed estimator has the nested structure identical to the NFXP estimator of Rust (1987): on the outer loop the likelihood function is maximized with respect to the structural parameters, while each evaluation of the likelihood requires the solution of the dynamic problem, forming the inner loop. IRS2016 provide the RLS algorithm for computing the full set of MPE in a class of directional dynamic games, which we use to solve the inner loop problem. It is due to potential multiplicity that the estimator has *an additional maximum* over $\mathscr{E}(\theta)^M$ or $\mathscr{E}(\theta)$ in (21).

Next, we make the following useful observation. The market-specific likelihood function $\mathscr{L}_T(m,\theta,e)$ can be defined in two equivalent ways: first, as a sum over the time periods $t=1,\dots,T$ that the players are observed to take actions a in states x in market m. Alternatively, it can be computed as a sum

[8] Just the computation of the full set of equilibria in dynamic games has been a major obstacle in the literature with only few exceptions (Borkovsky, Doraszelski and Kryukov, 2010; Herings and Peeters, 2004; Besanko, Doraszelski, Kryukov and Satterthwaite, 2010; Judd, Renner and Schmedders, 2012; Doraszelski and Judd, 2012).

over the state space of the game — as turns out to be very useful for the development of the efficient implementation of the NRLS estimator. Let $n_i(\alpha,x,m)=\sum_{t=1}^{T}\mathbb{1}\{a_{imt}=\alpha,x_{mt}=x\}$ denote the count of observations of player i taking action $\alpha\in A$ in state $x\in X$ in market $m\in\{1,\dots,M\}$ in the observed sample $\{(x_{mt},a_{mt})\}_{m=1,t=1}^{M,T}$. Then we have the following two equivalent expressions for the market-specific likelihood function $\mathscr{L}_T(m,\theta,e)$:

$$\mathscr{L}_T(m,\theta,e)=\frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{N}\log P_i(a_{imt}|x_{mt};e,\theta)=\frac{1}{T}\sum_{x\in X}\sum_{\alpha\in A}\sum_{i=1}^{N}n_i(\alpha,x,m)\log P_i(\alpha|x;e,\theta). \quad (22)$$

The role of second representation in (22) is more than just convenience: it is the one that allows us to exploit the directional structure of the game in the computation of choice probabilities $P_i(\alpha|x;e,\theta)$ characterizing different equilibria $e\in\mathscr{E}(\theta)$, and eventually avoid computing the equilibria that do not appear promising in the maximization of the likelihood of the sample. To achieve this, we employ the branch-and-bound (BnB) algorithm (Land and Doig, 1960) which is a classic solution approach for the integer programming problems, but fits beautifully for our problem of maximizing the likelihood function over a finite set of equilibria. Two key ingredients of the BnB algorithm are the tree structure of the search space and the bounding function that allows to prune the search tree. We describe these ingredients in turn in the next two sections.

3.3 Recursive lexicographic search algorithm (RLS)

The recursive lexicographic search (RLS) algorithm of IRS2016 finds all MPE of a directional dynamic game by decomposing this problem into a series of small subproblems, the *stage games*, associated with the substages of the state space introduced in Section 2.2, and by traversing the lexicographically ordered list of combinations of their solutions using a specially designed *successor function*. In this section we restate the decomposition result of IRS2016 in the notation of the present paper, and give a new interpretation of their algorithm: RLS is a depth-first traversal (Tarjan, 1972) of a particular tree, the *RLS tree*, whose nodes are the stage game equilibria and whose root-to-leaf paths are the MPE of the game. Beyond making the algorithm transparent, the tree interpretation is what allows us to combine the computation of $\mathscr{E}(\theta)$ with the maximization of the likelihood over it in Section 3.4.

Given the decomposition $X=D\times X'$ of the state space of the game, let the index $\tau\in\{1,\dots,|D|\}$ enumerate the substages $B_d=\{d\}\times X'$ along the total order $\trianglelefteq$ on D that is guaranteed by Proposition 2.1 to exist in any directional game. Denote by d_τ the element of D corresponding to substage τ, so that the substage of the game indexed τ is $B_{d_\tau}=\{d_\tau\}\times X'$. Let the direction of the τ index be defined such that the first substage $\tau=1$ corresponds to the minimal element of D in the order $\trianglelefteq$, and the last substage $\tau = |D| \equiv \bar{\tau}$ corresponds to the maximal element of D. In other words, $\tau = \bar{\tau}$ is the *end game* where no further transitions are possible except for the non-directional transitions within substage $B_{d_{\bar{\tau}}}$. The existence of such an index is a purely graph-theoretic fact: because directionality renders the precedence relation between the substages acyclic—equivalently, by Proposition 2.1, the transition matrices are block-triangular (8)—the enumeration τ is nothing but a topological sort of the maximal transition graph $\bar{\mathscr{D}}$ (Kahn, 1962).[9]

Fix $\theta \in \Theta$ and consider solving the game backward along this order. Suppose that at every substage $\tau' > \tau$ one equilibrium has already been computed and selected, recursively for $\tau' = \bar{\tau}, \dots, \tau + 1$, so that the selected choice probabilities and the value functions they imply are available at every state in $X_{>\tau} = \bigcup_{\tau' > \tau} B_{d_{\tau'}}$. By block-triangularity, from the states of B_{d_τ} the game can only transition within B_{d_τ} itself or into $X_{>\tau}$, so the equilibrium conditions (3)–(5) restricted to B_{d_τ} contain no unknowns outside the substage: they form a self-contained system of nonlinear equations in the choice probabilities and values of all N players at the $|X'|$ points of the substage, in which the selected continuation values enter as known constants. We call this system—the fixed point condition $P = \Lambda(P;\theta)$ of Section 2.1 restricted to B_{d_τ}—the *stage game* at τ, and call its solutions *stage equilibria*: profiles of choice probabilities that are mutual best responses at every point of the substage, given that play reverts to the selected equilibria once the state leaves it. At least one stage equilibrium exists by the same Brouwer fixed point argument that guarantees $\mathscr{E}(\theta) \neq \emptyset$, but it need not be unique—the leapfrogging stage games discussed below can have as many as five—and both the number of stage equilibria and their values depend on the equilibria selected at the substages $\tau' > \tau$ through the continuation values. The following proposition restates in the notation of the present paper the decomposition results of IRS2016 (their Theorems 2–4). It requires no separate proof: by Lemma A.1 their constructions apply to the directional games of Definition 3, and the recursion over τ is the serialization of their stage-by-stage recursion, which leaves every stage game unchanged because the substages within one of their stages do not communicate.

Proposition 3.4 (Decomposition of the equilibrium set, IRS2016)**.** *For any $\theta \in \Theta$: (i) every profile of choice probabilities on X constructed by selecting, recursively for $\tau = \bar{\tau}, \bar{\tau}-1, \dots, 1$, one stage equilibrium of the stage game at τ induced by the previously selected stage equilibria, is an MPE of the game, that is, an element of $\mathscr{E}(\theta)$; (ii) conversely, every $e \in \mathscr{E}(\theta)$ arises from such a recursive selection: for every τ, the restriction of the choice probabilities $P_i(a_i|x;e,\theta)$ to $x \in B_{d_\tau}$ is a stage equilibrium of the stage game at τ induced by the continuation values implied by e on $X_{>\tau}$.*

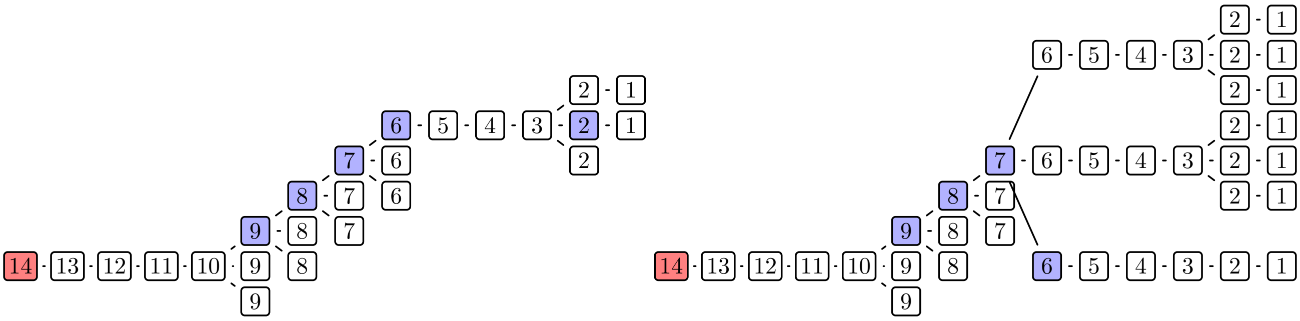
[9] IRS2016 arrange the recursion in two levels: the *stages* of the game, obtained by recursively stripping the terminal layer off the graph of feasible transitions between the elements of D, and the substages within a stage, which cannot communicate under any strategy profile and may therefore be taken in any order. Enumerating the substages stage by stage produces an index of the kind used here, and it is this serialization of their recursion that we work with below.

Part (i) generalizes finite horizon backward induction over time to backward induction along the total order of substages: a single backward pass $\tau = \bar{\tau}, \dots, 1$ that solves each stage game and selects one of its equilibria—the *state recursion algorithm* of IRS2016—produces an MPE without ever solving a fixed point problem on the whole state space. Part (ii) delimits what any full solution method must do: the enumeration of $\mathscr{E}(\theta)$ is complete if and only if every stage game induced by every recursive selection is solved for *all* of its stage equilibria. No substage can be skipped and no stage equilibrium overlooked, because an equilibrium missed at a single substage removes from the enumeration every MPE that extends it.[10] With degenerate private shocks, $\eta = 0$, the leapfrogging stage games can be solved exactly and non-iteratively for all of their equilibria (Theorems 6 and 7 of IRS2016): the substages of the simultaneous move version are single states, so the stage game reduces to a pair of coupled equations in the two firms' investment probabilities whose solutions are the intersections of their best response functions and number one, three, or five, while in the alternating move version the substages consist of the two states of $X' = \{1, 2\}$ and the count is one or three. For $\eta > 0$ we rely on the stage game solution algorithms for the two versions detailed in the Online Appendix, which locate the stage equilibria up to the resolution of a piecewise linear approximation of the best response functions.

The recursive selections of Proposition 3.4 are naturally arranged into a tree, which we call the RLS tree and which is illustrated in Figure 2. Its levels correspond to the substages, running from the end game $\tau = \bar{\tau}$ at the root to the initial substage $\tau = 1$ at the leaves, and its nodes at level τ are the stage equilibria of the stage games at substage τ: the root is a stage equilibrium of the end game, and the children of a node at level τ are the stage equilibria at substage $\tau - 1$ induced by the selections along the path from the root to that node (if the end game itself admits several stage equilibria, each of them is a root and the RLS tree is a forest, traversed one component at a time). A node branches exactly when the stage game one level down has multiple equilibria. By Proposition 3.4 the complete root-to-leaf paths are exactly the MPE of the game, so, provided every stage game is solved for all of its equilibria, the leaves of the RLS tree are in one-to-one correspondence with $\mathscr{E}(\theta)$; every leaf then lies at level 1, and the tree is finite by Assumption B(i). Unlike in standard applications of graph traversal, however, the RLS tree is not known in advance: the children of a node—the number and the values of the stage equilibria one level down—are revealed only by solving the corresponding stage game, so the tree has to be discovered in the course of its own traversal. The two panels of Figure 2 are snapshots of this discovery at two steps of the traversal, for the simultaneous move leapfrogging game with $n = 3$, whose $|X| = 14$ singleton substages give the tree 14 levels. Each box is a stage equilibrium

[10] If the stage game solver finds only some of the stage equilibria, the search described below still terminates and returns a subset of $\mathscr{E}(\theta)$ (Corollary 5.1 of IRS2016), over which the inner maximization in (21) is then taken.

Figure 2: Example of partially traversed RLS tree.



Notes: The figure shows two different intermediate steps of the traversal algorithm with a partially extended RLS tree. Substages of the leapfrogging game with costs $c_i \in \{0,1,2\}$ discretized into $n = 3$ levels are indexed using the same total order of substages as in Figure 1, from the absorbing state $(c_1, c_2, c) = (0,0,0)$ indexed $\tau = 14$ to the apex $(c_1, c_2, c) = (2,2,2)$ indexed $\tau = 1$.

labeled by the index τ of its substage, with the root ($\tau = \bar{\tau} = 14$, the absorbing state) on the left and the leaves ($\tau = 1$, the apex) on the right; boxes stacked vertically are the several stage equilibria found at one substage, and the shaded boxes mark the substages at which the stage game turned out to have more than one equilibrium, that is, the nodes at which the tree branches. The later snapshot has more of the tree uncovered, but even there the parts of it that the traversal has not yet reached remain unknown.

The successor function of IRS2016 walks this tree depth first, discovering it as it goes, and we state this identification without a formal proof. It has an immediate implication for the cost of full enumeration: because all MPE that share a prefix of selections share the corresponding computations, each stage game arising in the tree is solved only once, so the number of calls to the stage game solver is the number of nodes of the RLS tree rather than the number of its paths $|\mathscr{E}(\theta)|$ times its depth $\bar{\tau}$. Even so, complete traversal remains the computational bottleneck of the estimator when multiplicity is severe: the number of leaves in the leapfrogging example can run into hundreds of millions (Section 2.3). The inner problem in (21), however, requires not the whole of $\mathscr{E}(\theta)$ but only its most likely elements, and the RLS tree is exactly the tree-structured search space called for by the branch and bound method. Section 3.4 supplies its second ingredient, the bounding function, and shows how the depth-first traversal can be pruned using the data, delivering the exact maximizer of the likelihood while discovering only a fraction of the tree.

3.4 Inner loop of the NRLS estimator: branch and bound algorithm

The computational burden of the full solution estimator is large and hard to predict, because the number of MPE—and with it the size of the RLS tree—varies with the trial value of θ over the course of the outer maximization in (21). This section develops the bounding rule that lets the depth-first traversal of Section 3.3 deliver the inner maximum while leaving most of the tree unexplored.

Branch and bound originated with Land and Doig (1960), who proposed solving linear programs with integer variables by relaxation and recursive subdivision; Little, Murty, Sweeney and Karel (1963) coined the name in their algorithm for the traveling salesman problem, and Lawler and Wood (1966) distilled the general scheme: recursively partition the feasible set into a tree of subsets (branch), compute a bound on the best objective value attainable within each subset (bound), and discard the subsets whose bound cannot beat the best complete solution found so far (prune, or *fathom* in the classical terminology). The method remains actively developed in computer science and operations research; see Morrison, Jacobson, Sauppe and Sewell (2016) for a modern survey organized around the design of its three components: the search strategy, the branching rule, and the pruning (bounding) rule.

All three components are readily available in our problem. Consider first the single equilibrium case of (21), where the inner problem is the maximization of $\frac{1}{M}\sum_{m=1}^{M}\mathscr{L}_T(m,\theta,e)$ over the finite set $\mathscr{E}(\theta)$. The RLS tree provides the branching: a node v at level τ represents the subset $\mathscr{E}_v(\theta)\subseteq\mathscr{E}(\theta)$ of the MPE whose recursive selections pass through v, the children of v partition $\mathscr{E}_v(\theta)$ according to the stage equilibrium selected at the next substage, and each leaf is a singleton. The search strategy is the depth-first traversal, which maintains the *record* (the incumbent, in the branch and bound terminology): the highest likelihood among the equilibria completed so far. What remains is the bounding rule: an upper bound on the likelihood attainable by any equilibrium in $\mathscr{E}_v(\theta)$, computable at the moment v is first reached. Whenever the bound falls below the record, the subtree rooted at v is pruned: none of the stage games in it ever needs to be solved.

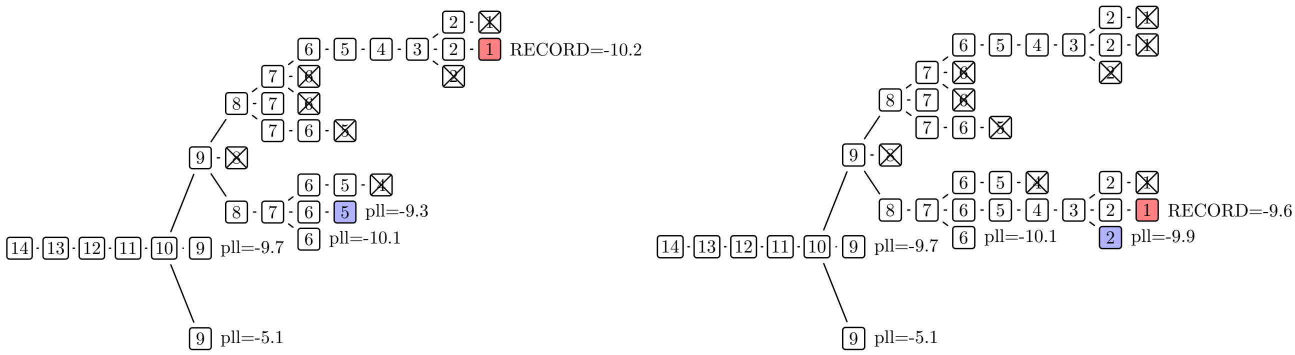
The building block of the bound is the likelihood of the data that a node already determines. Pooling the counts of (22) across markets, $n_i(\alpha,x)=\sum_{m=1}^{M}n_i(\alpha,x,m)$, the objective of the inner problem is

$$\mathscr{L}_{MT}(\theta,e)=\frac{1}{M}\sum_{m=1}^{M}\mathscr{L}_T(m,\theta,e)=\frac{1}{MT}\sum_{x\in X}\sum_{\alpha\in A}\sum_{i=1}^{N}n_i(\alpha,x)\log P_i(\alpha|x;e,\theta). \tag{23}$$

By the construction of the RLS tree, all equilibria in $\mathscr{E}_v(\theta)$ share the stage equilibria along the path to v, and thus share the choice probabilities at every state in $X_{\geq\tau}=B_{d_\tau}\cup X_{>\tau}$; denote these $P_i(\alpha|x;v,\theta)$. The *partial likelihood* of node v is the part of (23) that the node determines,

$$\mathscr{L}_{MT}(\theta,v)=\frac{1}{MT}\sum_{x\in X_{\geq\tau}}\sum_{\alpha\in A}\sum_{i=1}^{N}n_i(\alpha,x)\log P_i(\alpha|x;v,\theta), \tag{24}$$

Figure 3: Branch and bound traversal of the RLS tree.



Notes: The figure shows two intermediate steps of the branch and bound traversal of the RLS tree for the simultaneous move leapfrogging game with $n = 3$, with substages indexed as in Figure 2. Boxes are the discovered stage equilibria labeled by the substage index τ; hypothetical partial log-likelihood values (pll) annotate the frontier nodes, and crossed boxes mark the abandoned parts of the tree: pruned subtrees and completed branches overtaken by the record. Left panel: two branches have been fully extended, and the current log-likelihood record -10.2 is attained by the second of them (red box); the node shown in blue is extended next, since its partial likelihood -9.3 is still above the record. Right panel: five steps later the blue branch has been completed, establishing a new record of -9.6 (red box) and retiring the previous one (now crossed); the frontier node with pll of -9.9, shown in blue, falls below the new record and is pruned without solving any further stage games.

available as soon as the stage games along the path to v are solved. Because every term of (23) is nonpositive, the partial likelihood can only decrease as the traversal extends the path and more data are accounted for:

$$\mathscr{L}_{MT}(\theta, e) \leq \mathscr{L}_{MT}(\theta, v') \leq \mathscr{L}_{MT}(\theta, v) \leq 0 \tag{25}$$

for any node v, any descendant v' of v, and any $e \in \mathscr{E}_{v'}(\theta)$; at a leaf, where $X_{\geq 1} = X$, all data are accounted for and the partial likelihood equals the full likelihood (23) of the corresponding equilibrium. By (25), the partial likelihood is itself a valid bounding function, and pruning the branches whose partial likelihood falls below the record is the standard form of the branch and bound search. Figure 3 illustrates it on the RLS tree of Figure 2: the crossed boxes mark the pruned subtrees, and the annotated frontier nodes await extension or pruning.

The standard rule discards a branch only after its partial likelihood has already fallen below the record, and thus implicitly extrapolates the remaining likelihood contribution of the branch by zero—as if the model could fit the data at the remaining substages perfectly. The data, however, limit how well any model can possibly fit, and this limit is available before a single equilibrium is computed. For each state and player define the *nonparametric likelihood*

$$\ell_i^{np}(x) = \sum_{\alpha \in A} n_i(\alpha, x) \log \frac{n_i(\alpha, x)}{n_i(x)}, \qquad n_i(x) = \sum_{\alpha \in A} n_i(\alpha, x), \tag{26}$$

with the conventions $0 \log 0 = 0$ and $\ell_i^{np}(x) = 0$ whenever $n_i(x) = 0$. This is the maximum of $\sum_\alpha n_i(\alpha, x) \log q(\alpha)$ over all probability distributions q on A, attained at the empirical frequencies $q(\alpha) = n_i(\alpha, x)/n_i(x)$, so no choice probabilities, equilibrium or not, can fit the observations at x better. It depends neither on θ nor on the equilibrium, and is computed once from the data, before the model is ever solved. Replacing the zero extrapolation of the standard rule with the remaining nonparametric likelihood yields the bounding function

$$U(\nu; \theta) = \mathscr{L}_{MT}(\theta, \nu) + \frac{1}{MT} \sum_{x \in X_{<\tau}} \sum_{i=1}^{N} \ell_i^{np}(x), \qquad X_{<\tau} = X \setminus X_{\geq \tau}, \tag{27}$$

for a node ν at level τ.

Proposition 3.5 (Validity of the bounding rule)**.** *For any $\theta \in \Theta$ and any node ν of the RLS tree at level τ: (i) $\mathscr{L}_{MT}(\theta, e) \leq U(\nu; \theta) \leq \mathscr{L}_{MT}(\theta, \nu)$ for every $e \in \mathscr{E}_\nu(\theta)$; (ii) if ν is a leaf, then $U(\nu; \theta) = \mathscr{L}_{MT}(\theta, e)$ for the equilibrium e it represents.*

Proof. Any $e \in \mathscr{E}_\nu(\theta)$ shares the choice probabilities of ν on $X_{\geq \tau}$, so the difference between (23) and (24) is the sum of the terms over $x \in X_{<\tau}$ scaled by $\frac{1}{MT}$, and for each such x and each i, Gibbs' inequality (the nonnegativity of the Kullback–Leibler divergence) gives $\sum_\alpha n_i(\alpha, x) \log P_i(\alpha | x; e, \theta) \leq \ell_i^{np}(x)$. Summing and scaling yields the first inequality in (i); the second follows from $\ell_i^{np}(x) \leq 0$. At a leaf $X_{<1} = \emptyset$, so the sum in (27) is empty and (ii) reduces to the leaf case of (25). $\square$

Part (i) states that U is an admissible bounding function, weakly sharper than the partial likelihood of the standard rule; the two coincide when the remaining substages carry no observations or no variation in the observed choices. Part (ii) is the tightness at the leaves that the branch and bound method requires of a bounding function. Together they imply that the traversal that prunes every node with $U(\nu; \theta)$ below the record terminates with the record equal to $\max_{e \in \mathscr{E}(\theta)} \mathscr{L}_{MT}(\theta, e)$ and the recorded leaf equal to a maximizer:[11] the inner problem in (21) is solved *exactly*—the maximization over the discrete set $\mathscr{E}(\theta)$ involves no approximation—even though only a fraction of the tree is discovered and most elements of $\mathscr{E}(\theta)$ are never computed. Successive traversals within the outer loop of (21) also share information: the search at a new trial value of θ first extends the equilibrium

[11] Provided, as throughout, that every stage game is solved for all of its stage equilibria (Theorem 5 of IRS2016; cf. Section 3.3).

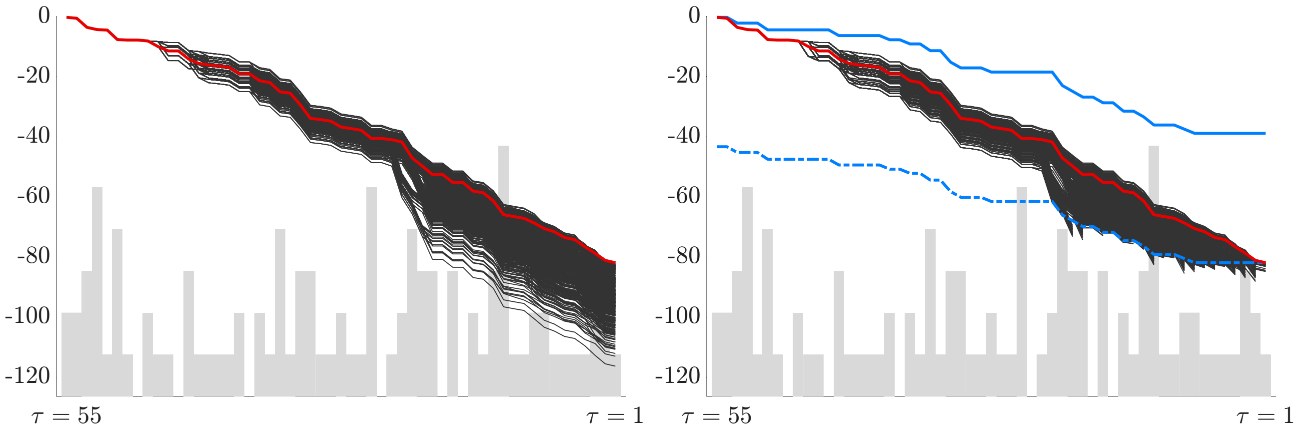
selection that attained the record at the preceding value, and the likelihood of the resulting complete equilibrium supplies a strong incumbent before any other branch is explored.

Figure 4 visualizes the resulting computational gains on simulated data. In the left panel, the fan of partial likelihood paths widens at every substage with multiple stage equilibria; the paths are flat across the states that carry no observations, as is any path at a state where its choice probabilities are (numerically) degenerate at the observed actions. In the right panel, the pruning threshold implied by (27) cuts branches shortly after they separate from the surviving ones: by Gibbs' inequality applied state by state, the nonparametric line declines the slowest, so a branch that has fallen behind can never catch up with the threshold, and the entire subtree below is pruned. Since the threshold is determined by the current record, the search benefits from finding high records early; extending the children in the order of their partial likelihood (a greedy refinement of the depth-first order) approaches the sharpest threshold shown in the figure.

The bounding rule sharpens as the sample grows. Figure 5 repeats the right panel of Figure 4 for five and ten times as many markets, and pruning strikes ever closer to the branching points. The reason is that the nonparametric likelihood is not merely a bound but a consistent estimate of the attainable likelihood: under correct specification, the gap between $\ell_i^{np}(x)$ and the corresponding term of (23) at the data generating equilibrium equals $n_i(x)$ times the Kullback–Leibler divergence of the empirical choice frequencies at x from the true choice probabilities, and remains bounded in probability as the cell counts grow. After the $\frac{1}{MT}$ normalization this gap vanishes, while the likelihood separation between distinct equilibria settles at positive constants, so inferior branches cross the threshold ever closer to the root. For the exact full solution estimator this turns the usual computational logic on its head: more data mean fewer computations, so long as the observations pooled under a common equilibrium grow with the sample, and it is precisely in large samples—where the statistical case for the full solution estimator is strongest—that its computational burden is lowest, opening the possibility of estimating substantially larger models.

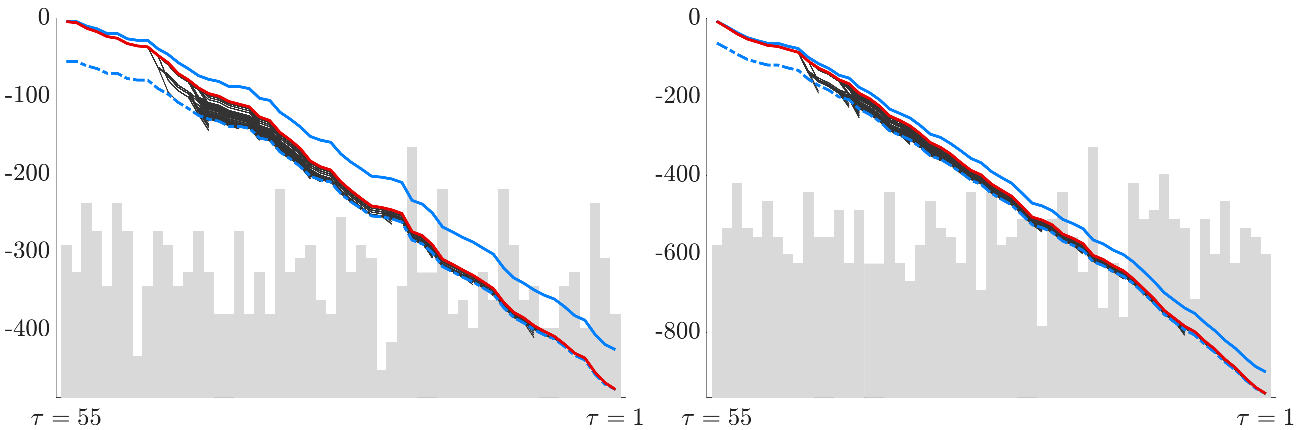
In the multiple equilibria case of (21) the inner problem separates into M market-specific maximizations over the same set $\mathscr{E}(\theta)$, and a single traversal serves them all: partial likelihoods, nonparametric terms and records are tracked market by market using the counts $n_i(\alpha, x, m)$, and a node is pruned once its market-specific bound falls below the corresponding record for every market. The recorded leaves then deliver the estimated profile $\hat{e}_{1:M}$ of (21). Section 4 documents the computational gains delivered by the branch and bound inner loop in practice, alongside the statistical performance of the NRLS estimator.

Figure 4: Partial log-likelihood along the RLS tree: full enumeration and branch and bound.



Notes: Both panels plot partial log-likelihood paths for the simultaneous move leapfrogging game with $n = 5$ ($|X| = 55$), on $M = 100$ markets simulated from one equilibrium. The horizontal axis orders the states (c_1, c_2, c) by substage τ, from the absorbing state at the root of the RLS tree ($\tau = \bar{\tau} = 55$, left) to the apex at its leaves ($\tau = 1$, right); the background histogram gives the distribution of the observations over the states. Left: full enumeration by RLS—every branch is extended to its leaf; each line is one MPE, its rightmost value that equilibrium's full log-likelihood, and the red line the maximal likelihood equilibrium. Right: the same dataset under the bounding rule (27), showing only the expanded branches. The solid blue line is the partial nonparametric log-likelihood; the dash-dotted blue line subtracts the remaining nonparametric log-likelihood from the record and is the pruning threshold, drawn at its sharpest using the final record. Branches are abandoned within a substage of crossing it, so the search is confined to the corridor between the two blue lines.

Figure 5: Effect of the sample size on the amount of computation.



Notes: Same setting as in the right panel of Figure 4, for $M = 500$ (left) and $M = 1000$ (right) markets simulated from the same model. The corridor between the partial nonparametric log-likelihood and the pruning threshold stays roughly constant in absolute width while the decline of the likelihood paths scales with the sample size, so branches are pruned almost immediately after branching.

4 Monte Carlo experiments

This section evaluates the finite-sample and computational performance of NRLS and compares it with commonly used estimators of dynamic discrete games. The experiments progress from a unique-equilibrium benchmark to designs with stable and unstable equilibria, discontinuities in the full-solution likelihood, and massive multiplicity. We then study the convergence properties of NRLS and allow different groups of markets to play different equilibria, including a design with 19,683 MPEs in the model and 23 unique equilibria played in the data. We first describe the competing estimators and then present the common experimental design and results.

4.1 Existing estimators

We use simplified notation of Sections 2.1 and 3.1. Let P and V denote the CCP and the value functions profile of the players, such that $V = \Gamma(P;\theta)$, and an MPE equilibrium satisfies $P = \Psi(P;\theta)$. The dependence on the structural parameter θ is made explicit. Denote the observed data $Z = \{a_{mt}, x_{mt}\}_{m,t}$. Assuming uniqueness of equilibrium and disregarding the likelihood of transition the conditional log-likelihood is given by $\mathscr{L}(Z,P) = \sum_{m=1}^{M}\sum_{t=1}^{T}\sum_{i=1}^{N}\log P_i(a_{imt}|x_{mt};\theta)$.

Nested pseudo likelihood (NPL)

The nested pseudo likelihood (NPL) estimator of AM2007 avoids repeatedly solving for an equilibrium at every trial value of θ.[12] Following their notation, for any CCP profile P, define $\tilde{\theta}_M(P) = \arg\max_\theta \mathscr{L}(Z,\Psi(P;\theta))$ and the sample NPL operator $\phi_M(P) = \Psi(P;\tilde{\theta}_M(P))$. Starting from a non-parametric estimate $\hat{P}^0$ of the CCPs, 1-NPL holds this preliminary CCP profile fixed when constructing the pseudo-likelihood and solves $\hat{\theta}^1 = \tilde{\theta}_M(\hat{P}^0)$.

NPL recursively updates both objects: $\hat{\theta}^k = \tilde{\theta}_M(\hat{P}^{k-1})$ and $\hat{P}^k = \Psi(\hat{P}^{k-1};\hat{\theta}^k) = \phi_M(\hat{P}^{k-1})$. A limit is a fixed point of the sample NPL operator, but it need not be the NPL estimator: when there are several sample fixed points, the NPL estimator is the one attaining the highest pseudo-likelihood. The distinction matters because the iteration can fail to converge or can converge to an inferior fixed point. Local convergence of the sample algorithm is governed by the Jacobian of ϕ_M, which differs from the fixed-θ equilibrium-stability diagnostic used below. These distinctions between the NPL estimator and the algorithm used to compute it are emphasized by Aguirregabiria and Marcoux (2021). In dynamic games, neither fixed-k nor converged NPL is generally asymptotically efficient.

[12] Related CCP estimators are described in Pakes, Ostrovsky and Berry (2007)

Efficient pseudo likelihood (EPL)

The efficient pseudo likelihood (EPL) estimator of Dearing and Blevins (2025) replaces NPL's policy update in CCP space with a Newton-type update in the space of choice-specific value functions. Let v stack the choice-specific values in (3). Define $G(v;\theta)$ as the difference between v and the choice-specific values implied by (θ,v), where v determines CCPs through (5), and those CCPs determine expected payoffs and transitions through (1). Thus, an equilibrium satisfies $G(v;\theta)=0$. Given the preceding iterate $(\hat{\theta}^{k-1},v^{k-1})$, EPL uses

$$\Upsilon_k(\theta)=v^{k-1}-\left[\nabla_v G(v^{k-1};\hat{\theta}^{k-1})\right]^{-1}G(v^{k-1};\theta) \tag{28}$$

and maximizes the likelihood of the CCPs implied by $\Upsilon_k(\theta)$. Starting from a suitable $\sqrt{M}$-consistent estimator, each fixed-k EPL estimator is asymptotically efficient under the regularity conditions in Dearing and Blevins (2025). Under their nonsingularity and uniqueness conditions, and when initialized sufficiently close to a regular maximum-likelihood estimate, iteration converges locally to the finite-sample MLE with probability approaching one. With linear utilities and logit errors, an EPL update consists of solving linear systems and a static multinomial choice problem. Its computational structure is therefore similar to NPL, although the larger linear systems can make an EPL iteration more expensive. We initialize EPL with the 1-NPL estimates; one update is denoted 1-EPL, whereas EPL denotes iteration to convergence. These guarantees are local: EPL does not search all equilibrium branches.

Mathematical programming with equilibrium constraints (MPEC)

Egesdal, Lai and Su (2015) formulate maximum likelihood estimation as a constrained optimization problem jointly over the structural parameters, integrated value functions, and CCPs. To describe their formulation let $\Xi^V(V,P;\theta)$ denote the value-function update on the right-hand side of the integrated Bellman equation (6). Let $\Xi^P(V,P;\theta)$ denote the corresponding probability update: choice-specific values are constructed from (θ,V,P) according to (3), and the implied CCPs are then obtained from (5).[13] They evaluate an augmented likelihood using the probabilities $\Xi^P(V,P;\theta)$. Because the probability constraint imposes $P=\Xi^P(V,P;\theta)$, that objective equals $\mathscr{L}(Z,P)$ on the feasible set. Their constrained estimator can therefore equivalently be written as

$$\max_{\theta,V,P}\mathscr{L}(Z,P)\ \text{ subject to }\ V=\Xi^V(V,P;\theta),\ P=\Xi^P(V,P;\theta). \tag{29}$$

[13] Egesdal, Lai and Su (2015) denote these component mappings by Ψ^V and Ψ^P. We use Ξ^V and Ξ^P to avoid confusing their notation with Ψ, which Section 2.1 uses, following AM2007, for the CCP-to-CCP policy-iteration operator.

We label this formulation MPEC-VP because both V and P remain optimization variables. The first constraint imposes the integrated Bellman equations, while the second imposes consistency between the CCPs and the choice-specific values generated by (θ, V, P). For any θ, the joint solutions to these constraints constitute the set of MPEs of the model.

Our MPEC-P implementation uses the Hotz–Miller representation and the policy-valuation operator Γ to eliminate V from the variables of the constrained optimization problem. Following AM2007, the integrated value function is uniquely determined as $V = \Gamma(P;\theta)$, and so $\Xi^P(\Gamma(P;\theta), P;\theta) = \Psi(P;\theta)$, leading to the alternative MPEC-P formulation

$$\max_{\theta,P} \mathscr{L}(Z,P) \text{ subject to } P = \Psi(P;\theta) \tag{30}$$

The two formulations differ only in their optimization variables and constraint representation: both have the same equilibrium CCPs, and at a *global* solution either one maximizes the likelihood jointly over θ and all MPEs admitted at that parameter, and so implements the full-solution maximum-likelihood estimator.

That equivalence concerns global solutions only. In practice, the equilibrium constraints define a nonconvex feasible set with potentially many equilibrium branches, so a local constrained optimizer can converge to a feasible MPE that does not maximize the likelihood over all MPEs. NPL and EPL are similarly dependent on their initial values and local mappings. By contrast, conditional on each evaluated value of θ, NRLS explicitly maximizes the likelihood over all equilibria, using branch-and-bound to avoid full enumeration. The implementation and initialization of all estimators is described next.

4.2 Monte Carlo design and implementation

All experiments use the leapfrogging investment game in Section 2.3 analyzed on IRS2018, including developing conditions for uniqueness of equilibria. We vary the discretization, investment cost, move structure, the number and the nature of the equilibrium played in the data to move from a unique-equilibrium benchmark to small and then massive multiplicity, and finally to the case of different equilibria played at different markets. Experiments A–E and G use 100 Monte Carlo replications; Experiment F uses the larger and sample-size-specific replication counts described below.

Within each replication, we estimate the preliminary CCPs $\hat{P}^0$ from empirical choice frequencies and compute 1-NPL first. NPL starts from $\hat{P}^0$, EPL starts from the 1-NPL estimates, and NRLS and both MPEC formulations are initialized at the 1-NPL estimates. MPEC-VP additionally initializes V at the integrated values implied by the preliminary CCPs through the Γ policy valuation operator. Thus,

Table 1: Monte Carlo experiments design

Experiment	n	States	k_1	MPEs	DGP MPEs	M	T
A: Unique equilibrium	4	30	3.50	1	1	1,000	5
B: Small multiplicity; stable DGP equilibrium	4	30	7.50	3	1	5,000	5
C: Small multiplicity; unstable DGP equilibrium	4	30	7.50	3	1	5,000	5
D: Discontinuous likelihood	5	55	3.50	9	1	1,000	5
E: Massive multiplicity	6	91	3.75	2,455	1	1,000	5
F: Convergence: smooth/discontinuous	4/5	30/55	7.50/3.50	3/9	1	50–5,000	5
G_1: Five equilibrium assignments	5	55	9.25	81	3 (5 groups)	5,000	5
G_2: Twenty-five equilibrium assignments	6	91	9.50	19,683	23 (25 groups)	5,000	10

Note: The DGP MPEs column counts distinct equilibria played in the data; the number of groups is shown when several groups share an equilibrium. The M column reports markets per group in Experiments G1 and G2 and total markets otherwise. Experiment F repeats the smooth-likelihood and discontinuous-likelihood designs from Experiments B and D over a grid of sample sizes; the slash-separated entries report the two designs. Stability of data generated equilibrium is based on the spectral radius of the Jacobian of the policy-iteration mapping Ψ.

NRLS and MPEC receive the same preliminary parameter estimate; the comparisons do not depend on the starting values.

To make the computational comparisons informative, we use analytical first derivatives and sparse matrix implementations for the high-dimensional model operators and constraints. For NPL, the analytical pseudo-likelihood score accounts for the dependence of $\Psi(P;\theta)$ on θ. EPL constructs the Jacobian of its equilibrium residual analytically for the Newton update. Both MPEC formulations supply analytical gradients of the objective and sparse analytical Jacobians of the equality constraints. The derivatives for MPEC-VP follow directly from its two component equilibrium equations. MPEC-P instead requires the total derivative of the reduced constraint $\Psi(P;\theta)-P=0$ through the Hotz–Miller policy valuation; its Jacobian contains $\partial\Psi(P;\theta)/\partial\theta'$ and $\partial\Psi(P;\theta)/\partial P'-I$. Eliminating V therefore reduces the dimension of the constrained problem but makes construction of its constraint derivatives more involved. Within each experiment, all methods were run on the same hardware, so their CPU times are directly comparable.

We summarize the Monte Carlo designs in Table 1. Experiments A–E each place one equilibrium in the data while progressively increasing the difficulty of equilibrium computation and selection. Experiment F repeats two of these designs over a grid of sample sizes to study the statistical and computational properties of NRLS. Experiment G then allows different groups of markets to play different equilibria and culminates in a game with 19,683 MPEs.

Besides standard Monte Carlo summaries of bias and dispersion, we report several measures of equilibrium-selection accuracy. The KL divergence is computed between the estimated and data-generating equilibria as the divergence between the distributions of trajectories induced by the two sets of CCPs. Online Appendix contains a detailed presentation of the fixed point algorithm we developed for this calculation. All reported averages for an estimator use the replications in which that estimator converged. Rather than reporting log-likelihood that can be non-comparable across estimators due to non-converged cases and change mechanically with sample size, we report the likelihood shortfall relative to the best valid likelihood found by any method in that replication. The reported equilibrium residual $\|\Psi(P;\theta) - P\|$ is the measure of whether the estimated parameter induces an equilibrium. The probability-space distance $\|P - P_0\|$ is the average across the players of the maximum absolute difference between CCPs over all states and actions.

4.3 Experiments with one equilibrium played in the data

Figure 6 and Table 2 present complementary evidence for Experiments A–E. Each panel of the figure uses one representative Monte Carlo sample to illustrate the likelihood geometry and the estimates returned by the competing methods. The light curves are the equilibrium-specific criteria $\mathscr{L}_{MT}(\theta, e)$, and their upper envelope is the single-equilibrium full-solution likelihood in the first line of (21), up to the common normalization discussed above. NRLS evaluates this envelope by solving the inner maximization over $e \in \mathscr{E}(\theta)$. The table then reports performance over all Monte Carlo replications. We discuss the graphical and Monte Carlo evidence together, experiment by experiment. The sequence deliberately moves beyond the unique or stable settings emphasized in much of the earlier Monte Carlo literature. Table 2 holds the columns fixed across experiments. The blank MPEC entries in Experiment E are substantive: even with analytical derivatives and sparse constraint Jacobians, the dimensions of P and, for MPEC-VP, V render the constrained problems computationally infeasible in that design.

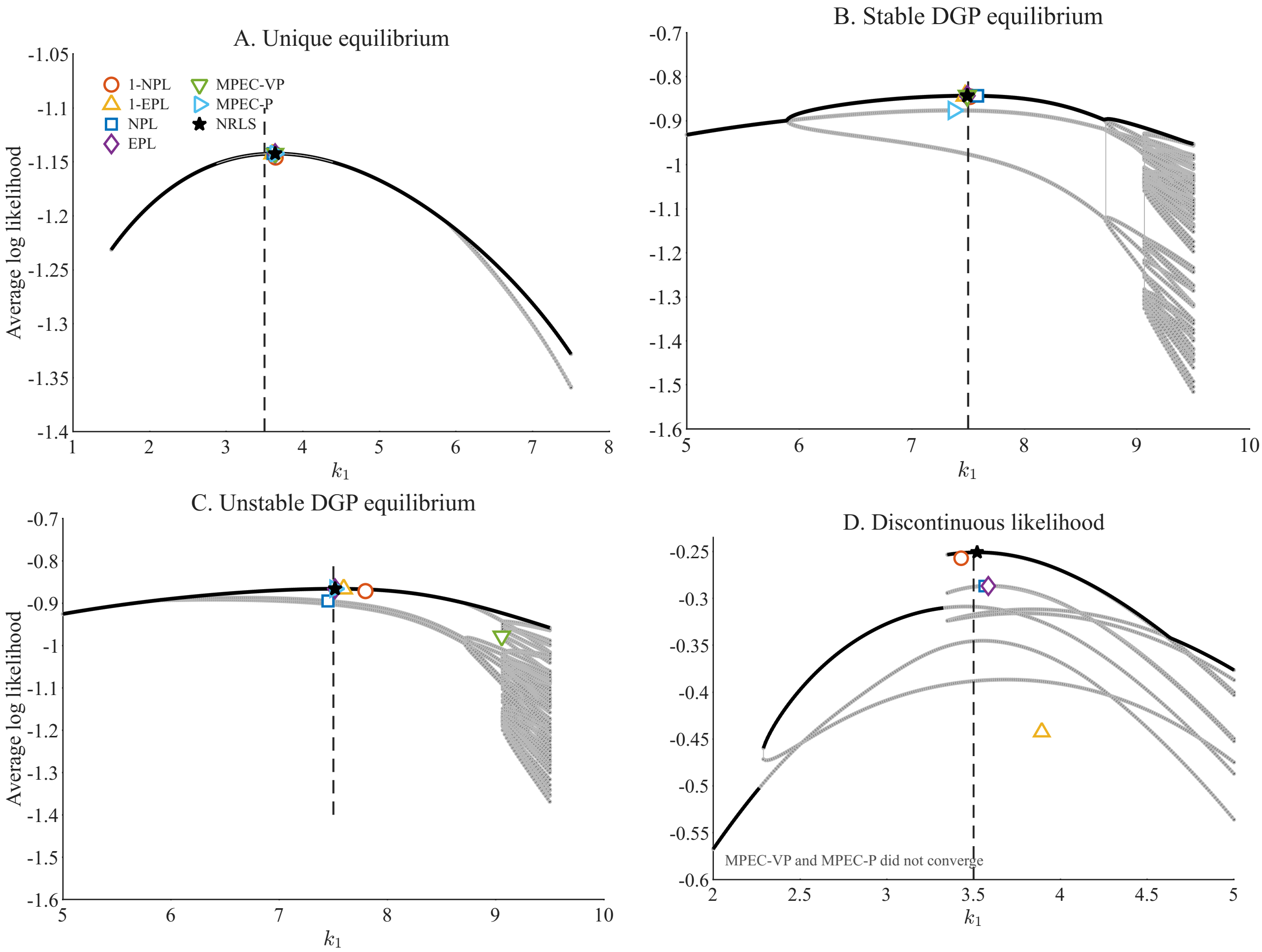
Experiment A: Unique equilibrium. The simultaneous-move game has $n = 4$ grid points in each cost dimension, 30 states, and investment cost $k_1 = 3.5$. Technological improvements have beta shape parameters $a = 1.8$ and $b = 0.4$, the idiosyncratic investment shocks have scale $\eta = 1$, and the common discount factor is $\beta = \exp(-0.05) = 0.9512$. Each replication contains $M = 1{,}000$ markets observed for $T = 5$ periods. Panel A of Figure 6 shows the single, smooth equilibrium branch, with the estimators clustered near its maximum. Table 2 confirms this agreement across replications. The 1-NPL estimator has the largest bias among the estimators that target the same structural parameter and retains a nonzero equilibrium residual, while iteration substantially reduces both. EPL, MPEC-VP, MPEC-P, and NRLS produce nearly identical parameter estimates and likelihoods. Thus, when equilibrium selection is absent, the full-solution and locally computed equilibrium estimators agree. This is the favorable benchmark emphasized in the earlier Monte Carlo literature: CCP estimators avoid repeated solution but can inherit error from preliminary CCPs, while iteration or constrained likelihood improves accuracy when equilibrium computation and selection do not bind, as in AM2007

Table 2: Monte Carlo results, Experiments A–E

	1-NPL	1-EPL	NPL	EPL	MPEC-VP	MPEC-P	NRLS
			Experiment A: Unique equilibrium				
Mean ($k_1^0 = 3.50$)	3.5279	3.4993	3.4971	3.4949	3.4949	3.4949	3.4949
Bias	0.0279	-0.0007	-0.0029	-0.0051	-0.0051	-0.0051	-0.0051
MCSD	0.1004	0.0709	0.0652	0.0704	0.0704	0.0708	0.0704
LL shortfall	–	–	-0.050	0.000	0.000	0.000	0.000
KL divergence	0.03254	0.00044	0.00021	0.00024	0.00024	0.00024	0.00024
$\|\|P-P_0\|\|$	0.11270	0.00995	0.00469	0.00495	0.00495	0.00500	0.00495
$\|\|\Psi(P;\theta)-P\|\|$	0.1618	0.0075	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
Converged (of 100)	100	100	100	100	100	99	100
CPU time (s)	0.01	0.03	2.00	0.13	0.27	0.23	3.81
			Experiment B: Small multiplicity; stable DGP equilibrium				
Mean ($k_1^0 = 7.50$)	7.5516	7.5052	7.4984	7.4992	7.6532	7.3512	7.4992
Bias	0.0516	0.0052	-0.0016	-0.0008	0.1532	-0.1488	-0.0008
MCSD	0.1788	0.0511	0.0606	0.0341	0.9974	0.4714	0.0341
LL shortfall	–	–	-1.2	0.000	-1062.7	-776.8	0.000
KL divergence	0.02557	0.00159	0.00040	0.00013	0.23536	0.16051	0.00013
$\|\|P-P_0\|\|$	0.11085	0.01749	0.00490	0.00280	0.17466	0.20957	0.00280
$\|\|\Psi(P;\theta)-P\|\|$	0.1709	0.0157	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
Converged (of 100)	100	100	100	100	98	97	100
CPU time (s)	0.01	0.03	1.58	0.12	0.76	1.23	4.22
			Experiment C: Small multiplicity; unstable DGP equilibrium				
Mean ($k_1^0 = 7.50$)	7.5424	7.4553	7.3928	7.4804	7.7313	7.6310	7.5018
Bias	0.0424	-0.0447	-0.1072	-0.0196	0.2313	0.1310	0.0018
MCSD	0.1714	0.2221	0.0561	0.1580	0.7299	0.8987	0.0382
LL shortfall	–	–	-765.2	-11.4	-502.1	-920.6	0.000
KL divergence	0.02271	0.00784	0.15996	0.00257	0.11452	0.20182	0.00012
$\|\|P-P_0\|\|$	0.09757	0.02547	0.20709	0.00619	0.03860	0.02504	0.00307
$\|\|\Psi(P;\theta)-P\|\|$	0.1601	0.0149	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
Converged (of 100)	100	100	18	100	99	98	100
CPU time (s)	0.01	0.03	1.95	0.14	0.61	0.70	3.88
			Experiment D: Discontinuous likelihood				
Mean ($k_1^0 = 3.50$)	3.4974	3.6219	3.5514	3.6477	3.6594	3.6703	3.5021
Bias	-0.0026	0.1219	0.0514	0.1477	0.1594	0.1703	0.0021
MCSD	0.1400	0.2117	0.0713	0.1290	0.1269	0.1158	0.0325
LL shortfall	–	–	-219.4	-222.1	-271.0	-267.5	0.000
KL divergence	0.01512	0.05395	0.04889	0.04495	0.04102	0.04078	0.00016
$\|\|P-P_0\|\|$	0.62850	0.77118	0.86124	0.83062	0.66562	0.65879	0.01610
$\|\|\Psi(P;\theta)-P\|\|$	0.7638	0.3628	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	1.5×10^{-6}
Converged (of 100)	100	100	100	100	28	27	100
CPU time (s)	0.02	0.04	0.85	0.22	2.30	0.52	22.0
			Experiment E: Massive multiplicity				
Mean ($k_1^0 = 3.75$)	3.7096	2.7906	3.7127	3.7891	–	–	3.7424
Bias	-0.0404	-0.9594	-0.0373	0.0391	–	–	-0.0076
MCSD	0.1109	0.7214	0.0681	0.4072	–	–	0.0303
LL shortfall	–	–	-66.5	-467.6	–	–	0.000
KL divergence	∞	∞	14.075	1.22×10^{4}	–	–	0.32429
$\|\|P-P_0\|\|$	0.82204	0.87083	0.65580	0.79241	–	–	0.07454
$\|\|\Psi(P;\theta)-P\|\|$	0.9636	0.8095	$<10^{-6}$	$<10^{-6}$	–	–	6.1×10^{-6}
Converged (of 100)	100	100	18	68	–	–	100
CPU time (s)	0.16	0.39	11.3	4.01	–	–	4.73

Note: Entries are averages over the replications in which the indicated estimator converged. For replication r, log-likelihood shortfall is $\mathscr{L}_{jr} - \max_{\ell\in\mathscr{A}_r}\mathscr{L}_{\ell r} \le 0$, where $\mathscr{A}_r$ contains the equilibrium estimators with a valid result in that replication. Thus NRLS is not normalized to zero. The reported KL divergence is accumulated recursively until absorption and is not normalized by expected path length, so it should be compared across estimators within a design rather than across designs. The CCP distance averages the two player-specific maximum absolute discrepancies from the data-generating CCPs, and the equilibrium residual is measured in probability space.

Figure 6: Likelihood correspondences in Experiments A–D



Note: Light gray curves are likelihood profiles conditional on individual equilibria, the black curve is the full-solution likelihood envelope, and the dashed vertical line gives the data-generating parameter. Estimator symbols are identified in panel A. Panels B and C use the same model but differ in whether the stable or unstable equilibrium generates the data. Crossed gray symbols and the accompanying annotation identify algorithms that did not converge. Panel D illustrates a change in the likelihood-maximizing equilibrium branch near the true parameter. The likelihood correspondence is reconstructed by computing all equilibria at a dense grid of parameter values and evaluating the likelihood at each equilibrium. Equilibria at the neighboring grid points which are closest in terms of the two-sided average KL divergence are then connected to form the branches.

and in Pakes, Ostrovsky and Berry (2007), Egesdal, Lai and Su (2015), and Dearing and Blevins (2025).

Experiment B: Small multiplicity with a stable equilibrium in the data. Increasing the investment cost to $k_1 = 7.5$ produces three MPEs at the true parameter. The data come from equilibrium 1, which is stable under policy iteration, with $\rho_\Psi(P^\star;\theta_0) = 0.77807$. Each replication contains $M = 5{,}000$ markets observed for $T = 5$ periods. Panel B displays the upper envelope of the equilibrium-specific likelihood profiles and shows how a local method can end on a branch below it. The large sample yields accurate empirical CCPs and hence favorable starting values for all local methods. NPL, EPL, and NRLS consequently estimate k_1 precisely, consistent with the favorable performance of sequential estimators in the stable-equilibrium designs of AM2007 and Dearing and Blevins (2025). Nevertheless,

Table 2 shows that both MPEC implementations sometimes converge to other feasible branches: their equilibrium residuals are essentially zero, but their likelihood shortfalls, KL divergences, bias, and dispersion are much larger. EPL and NRLS attain the best likelihoods in this design, while NPL is close but not identical. A negligible equilibrium residual therefore certifies that a reported policy is an equilibrium, not that it is the equilibrium attaining the global maximum of the likelihood.

This result adds a distinct dimension to the constrained-optimization evidence in Egesdal, Lai and Su (2015). In the presence of several feasible equilibrium branches, MPEC implements the full-solution MLE only if the constrained optimizer finds the global solution. Both MPEC formulations start from the same informative 1-NPL estimate as NRLS in this large sample, yet they sometimes end on inferior branches.

Experiment C: Small multiplicity with an unstable equilibrium in the data. The design holds the parameterization, state space, sample size, and set of three MPEs from Experiment B fixed, but generates the data from equilibrium 2. This equilibrium is unstable under policy iteration, with $\rho_\Psi(P^\star;\theta_0) = 1.2111$. The comparison therefore isolates the role of equilibrium stability. Panel C shows that policy iteration can be locally repelled from the statistically relevant branch. It also displays a distant MPEC-VP estimate, illustrating that satisfying the constraints can still leave a local optimizer on an inferior branch. The 1-NPL estimate need not lie on an equilibrium-specific profile because it does not impose the equilibrium restrictions. Across replications, NPL converges only 18 times and, when it does, has a substantial likelihood shortfall and KL divergence. This behavior accords with the local-contraction analysis of Kasahara and Shimotsu (2012) and the convergence-selection evidence of Aguirregabiria and Marcoux (2021). EPL is much more robust to policy-iteration instability, as predicted by Dearing and Blevins (2025), but its average likelihood remains below the best valid likelihood. Both MPEC variants usually return a feasible equilibrium, yet their likelihood shortfalls and dispersion reveal frequent convergence to inferior branches. NRLS evaluates the complete equilibrium correspondence and remains centered near the true parameter. The experiment therefore confirms EPL's contribution to local stability while identifying the complementary problem that a locally convergent method need not compare all equilibrium branches.

Experiment D: Discontinuous likelihood. The alternating-move game has $n = 5$, 55 states, $k_1 = 3.5$, and nine MPEs at the true parameter. The data are generated from an unstable equilibrium close to a parameter value at which the likelihood-maximizing branch changes. Panel D shows the resulting discontinuity, or likelihood cliff, in the full-solution likelihood. A local method that follows one branch can miss this discrete change in the optimizer over equilibria; in the illustrated replication, neither MPEC formulation converges. The Monte Carlo results show that NPL and EPL converge in all replications and satisfy the equilibrium restrictions after iteration, but their likelihood shortfalls and

KL divergences reveal frequent selection of equilibria far from the data-generating one. MPEC-VP and MPEC-P converge in only 28 and 27 replications, respectively. NRLS follows the full-solution likelihood envelope and has negligible bias relative to the other equilibrium estimators. The relevant difficulty is therefore not merely finding an equilibrium; it is tracking the globally maximizing equilibrium as that equilibrium changes discontinuously with the structural parameter.

Experiment F below compares this design with the smooth-likelihood design from Experiment B over a common grid of sample sizes. This comparison separates finite-sample distortions near the likelihood cliff from the large-sample behavior of the continuous parameter estimator and the discrete equilibrium classification.

Experiment E: Massive multiplicity. The simultaneous-move game now has $n = 6$, 91 states, $k_1 = 3.75$, and 2,455 MPEs at the true parameter, although only one equilibrium is played in the data. Exhaustively enumerating the equilibria by RLS at a single value of θ takes 10 minutes and 39 seconds. Repeating that enumeration at every trial parameter value would therefore be costly even before accounting for numerical derivatives or repeated Monte Carlo samples.

The sequential methods deteriorate sharply as multiplicity grows. NPL converges in only 18 replications and EPL in 68; when they converge, their likelihood shortfalls and probability distances show that they often select poorly fitting branches. MPEC is no longer computationally feasible because the CCP vector and equilibrium constraints are too large, notwithstanding analytical objective gradients and sparse analytical constraint Jacobians. MPEC-P removes V, but the smaller variable vector and analytically differentiated reduced constraint do not restore feasibility at this scale. This result complements Egesdal, Lai and Su (2015), who show that sparse analytical derivatives make constrained maximum likelihood practical in moderately sized games: sparsity remains valuable here, but does not eliminate the dimensional growth of P, V, and the equilibrium constraints. NRLS instead exploits directionality and likelihood bounds. It converges in all replications, has small parameter bias and dispersion, and takes 4.73 seconds on average; branch-and-bound obtains the full-solution likelihood without enumerating all 2,455 MPEs at each trial value.

Taken together, Experiments A–E separate two problems that are easily conflated. NPL and the methods developed to improve it address the stability and statistical efficiency of sequential estimation; EPL substantially improves local numerical stability. MPEC imposes the equilibrium restrictions directly and can implement maximum likelihood when a global constrained solution is obtained. Neither advance by itself resolves global selection among many MPEs, and the high-dimensional constrained formulation eventually becomes impractical. NRLS addresses this complementary problem for directional games: it evaluates the full-solution likelihood over the complete equilibrium correspondence without explicitly enumerating every MPE at every trial parameter value.

Experiment F: Convergence properties of NRLS.

We repeat the smooth-likelihood design from Experiment B and the discontinuous-likelihood design from Experiment D for

$$M \in \{50, 75, 100, 150, 200, 250, 350, 500, 750, 1000, 1250, 1500, 1750, 2000, 2250, 2500, 3500, 5000\},$$

holding $T = 5$ fixed. The smooth design has three MPEs and generates the data from the stable equilibrium used in Experiment B; the discontinuous design has nine MPEs and generates the data from the equilibrium near the likelihood cliff in Experiment D. Samples are nested within a replication, so increasing M adds markets to the same underlying simulated sample rather than drawing an unrelated dataset. We use 300 replications at every sample size and 1,000 replications at $M \in \{50, 250, 1000, 2500, 5000\}$, where the additional draws permit a more detailed examination of the finite-sample distributions.

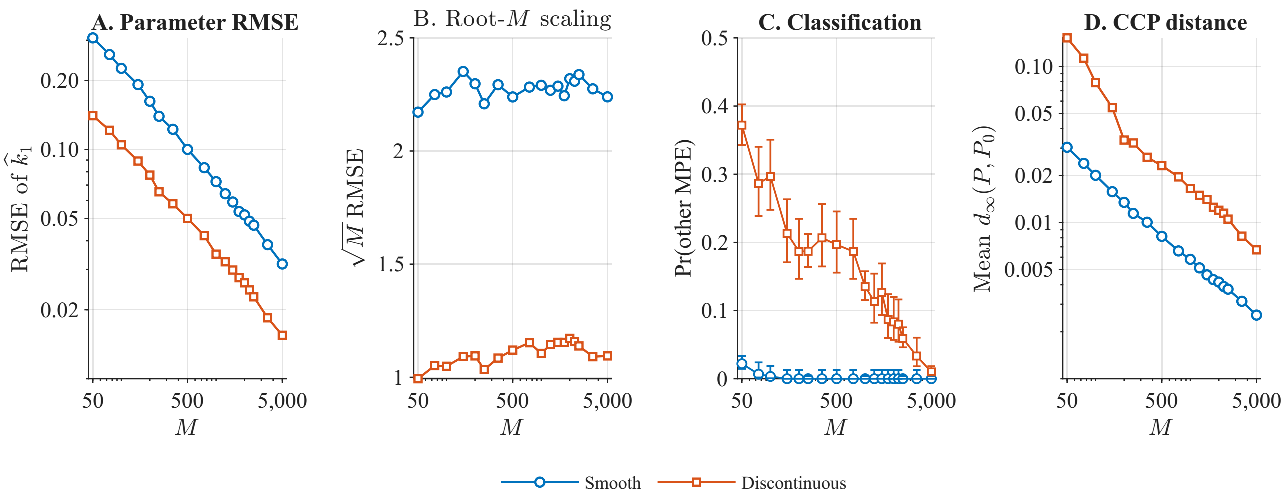
Growing M at fixed T is exactly the asymptotic sequence of Section 3.1, so the experiment inherits its indifference to the non-stationarity of the directional state process.

Table 3: Convergence properties of NRLS

M	R	Bias	RMSE	$\sqrt{M}$ RMSE	Other MPE (%)	Mean KL	Median KL	Mean $d_\infty(P, P_0)$
Smooth likelihood (Experiment B)								
50	1000	+0.0003	0.3071	2.17	2.2	1.41e-02	4.99e-03	0.0304
250	1000	+0.0008	0.1397	2.21	0.0	2.15e-03	9.39e-04	0.0114
1000	1000	-0.0039	0.0724	2.29	0.0	5.74e-04	2.68e-04	0.0058
2500	1000	-0.0027	0.0468	2.34	0.0	2.39e-04	1.02e-04	0.0037
5000	1000	-0.0017	0.0317	2.24	0.0	1.10e-04	4.77e-05	0.0026
Discontinuous likelihood (Experiment D)								
50	1000	+0.0245	0.1405	0.99	37.2	5.16e-03	1.37e-03	0.1524
250	1000	+0.0034	0.0654	1.03	18.7	5.41e-04	2.81e-04	0.0324
1000	1000	+0.0029	0.0350	1.11	13.5	1.57e-04	6.72e-05	0.0165
2500	1000	+0.0018	0.0228	1.14	5.9	6.76e-05	2.34e-05	0.0105
5000	1000	+0.0008	0.0155	1.09	1.0	2.49e-05	9.69e-06	0.0067

Note: R is the number of successful Monte Carlo replications. Other MPE is the percentage of classifiable replications in which NRLS selects an equilibrium branch other than the continuation of the data-generating branch. KL is the recursively accumulated divergence between the path distributions induced by the selected and data-generating equilibria. It is not normalized by expected path length and should therefore be compared across sample sizes within a design, not across the two designs. The probability-space distance is $d_\infty(P, P_0) = \frac{1}{2}\sum_{i=1}^{2} \|P_i - P_{i0}\|_\infty$.

Figure 7: Statistical convergence of NRLS



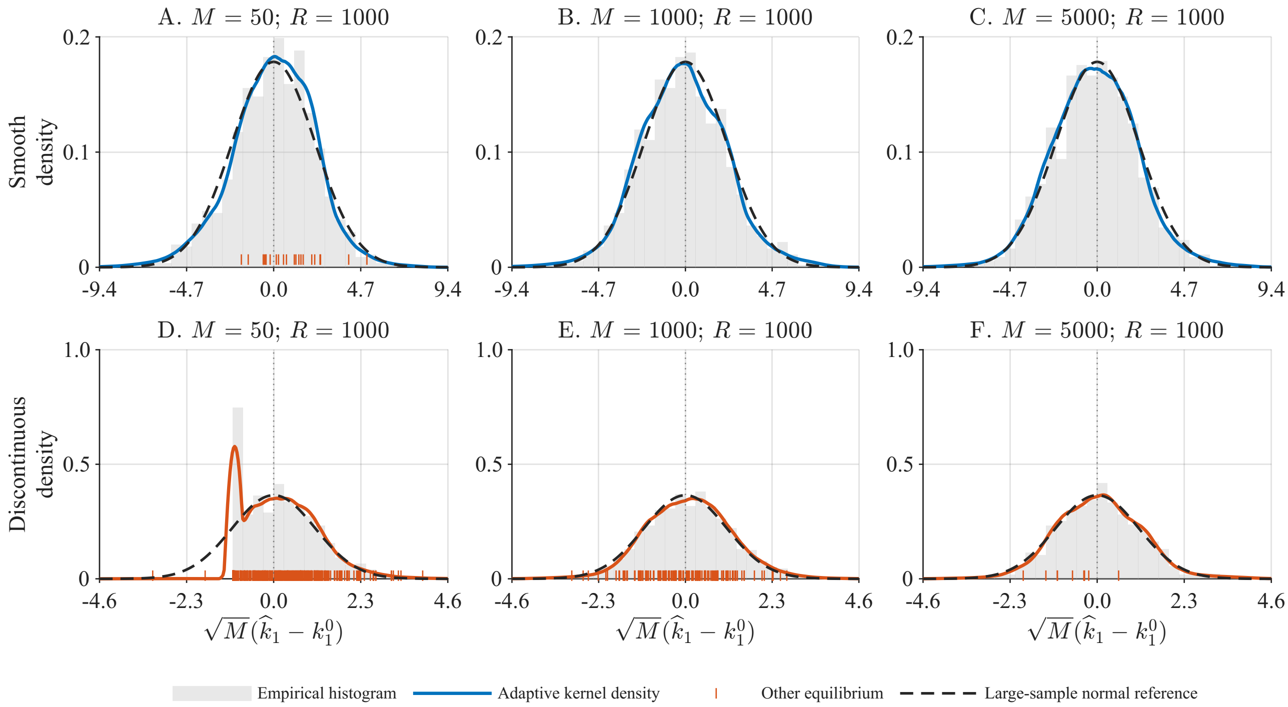
Note: The two designs repeat Experiments B and D. Panels A–D report the RMSE of $\hat{k}_1$, its root-M scaling, the probability of selecting another MPE, and $d_\infty(P, P_0) = \frac{1}{2}\sum_i \|P_i - P_{i0}\|_\infty$, respectively. Panel C gives 95 percent Wilson intervals.

Figure 7 separates convergence of the continuous parameter from classification of the discrete equilibrium branch. In both designs, the RMSE declines approximately linearly on logarithmic scales, while $\sqrt{M}$ times the RMSE is comparatively stable. The parameter estimates are therefore consistent with the root-M rate over the sample sizes considered. Equilibrium classification is already reliable in small samples under the smooth likelihood. Near the likelihood cliff, NRLS selects another equilibrium more frequently at the smallest sample sizes, but the probability falls sharply as the sample grows and approaches zero well before the continuous parameter uncertainty disappears. The CCP distance declines smoothly in both designs.

These patterns correspond to the two conclusions of the revised theory. Under the population separation condition in Assumption B(ii), Theorem 3.1 gives both $\hat{\theta} \to \theta_0$ and $\Pr(\hat{e} = e_0) \to 1$. Once the data-generating equilibrium is selected with probability approaching one, Theorem 3.2 establishes first-order equivalence to the oracle estimator that knows e_0 and, when θ_0 is in the interior of $\Theta(e_0)$, root-M asymptotic normality. The experiment illustrates both conclusions. The earlier disappearance of equilibrium-classification errors relative to continuous parameter uncertainty is an empirical finding; the theorems do not impose a relative rate for the two margins.

The two KL summaries in Table 3 add a distributional measure of equilibrium-selection accuracy. Both the mean and median KL divergence decline with M. At small sample sizes the mean can lie well above the median because a small number of replications produce substantially larger probability errors, often after selecting another equilibrium branch. The KL divergence is accumulated recursively until absorption and is not normalized by expected path length. It is therefore informative about convergence over M within each design, whereas the probability-space distance provides the more directly comparable scale across designs.

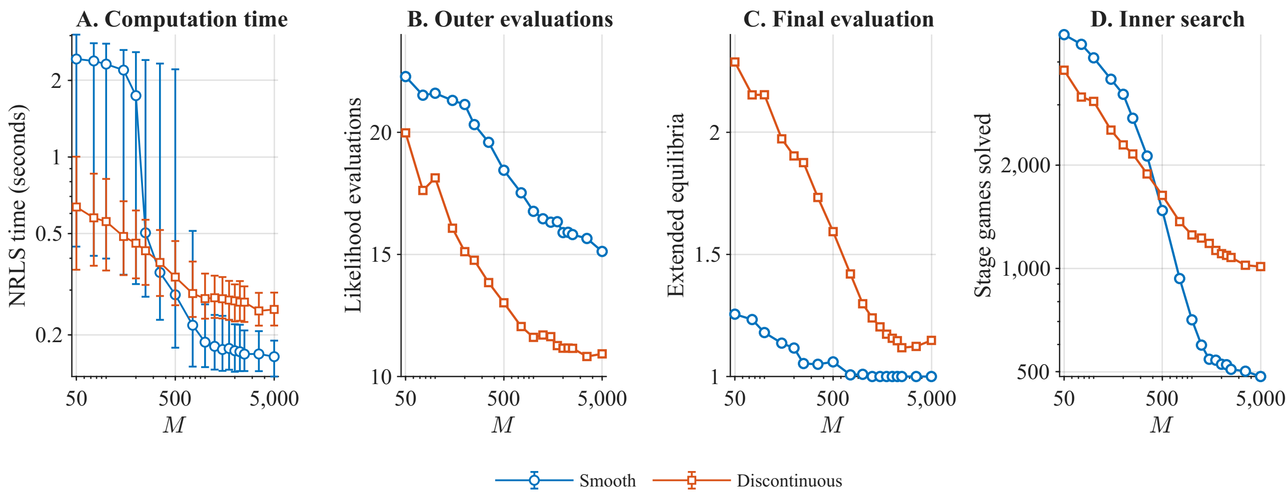
Figure 8: Finite-sample distributions of the NRLS estimator



Note: Histograms and kernel densities describe $\sqrt{M}(\hat{k}_1 - k_1^0)$. The dashed normal reference uses the variance at $M = 5{,}000$ within each design; orange marks identify another-MPE selections. Panel titles report successful replications, R.

Figure 8 shows why the distinction between parameter estimation and equilibrium classification matters. Under the smooth likelihood, the scaled parameter errors become approximately symmetric and close to the large-sample normal reference at relatively modest sample sizes. The discontinuous design exhibits a more irregular small-sample distribution, with replications selecting another equilibrium forming a visible component of that distribution. As M increases, these events become rare and the scaled distribution moves toward the normal reference, as predicted by the oracle result in Theorem 3.2. The likelihood cliff therefore creates a pronounced finite-sample distortion, but a change in the likelihood-maximizing branch does not by itself imply nonstandard asymptotics. The boundary case discussed after Theorem 3.2 arises when θ_0 lies on the boundary of the existence set $\Theta(e_0)$, contrary to Assumption C(i). The fixed data-generating parameter used here is separated from such a boundary and therefore does not establish a nonstandard limiting distribution.

Figure 9: Computational convergence of NRLS



Note: Panels A–D report median NRLS time (10th–90th percentile bars), outer likelihood evaluations, fully extended branches at the final evaluation, and cumulative stage games. Timing excludes 1-NPL and the validation audit; all runs use MATLAB R2025a on the same Apple M4 Max computer.

Figure 9 confirms the computational prediction of Section 3.4 in the sample. Although each likelihood evaluation processes more observations, median NRLS estimation time falls rather than rises with M: the number of outer likelihood evaluations declines, the number of equilibrium branches that must be fully extended at the final estimate approaches one, and the cumulative number of stage games solved over the estimation falls substantially. This is the empirical counterpart of the sharpening of the bounding rule (27) as the sample grows, and it explains the computational gains in Experiments E and G2 below.

4.4 Experiments with multiple equilibria played in the data

Results in this section are based on the Section 3.1 grouped estimator with fixed partition of markets into groups. NRLS solves the inner equilibrium-selection problem separately for each group while imposing the common structural parameter. The experiments condition on the partition used to generate the data. They therefore establish performance with known groups, not with unrestricted market-specific equilibrium fixed effects or latent group membership. The distinction matters for the asymptotics in Section 3.1.

The reported 1-NPL, 1-EPL, NPL, and EPL estimates are pooled benchmarks: they impose a common CCP profile and therefore do not implement the known-group likelihood above. Extending these methods to the grouped specification requires group-specific preliminary CCPs and policy updates. MPEC must additionally replicate the CCPs and equilibrium constraints for every group and, in MPEC-VP, the value functions as well. Analytical derivatives and sparsity mitigate but do not eliminate this dimensional growth. The experiments below therefore ask whether NRLS can estimate the common structural parameter and recover the group-specific equilibrium assignments when the partition is known. The pooled estimates illustrate the consequences of imposing a common CCP profile; they are not comparisons of numerical algorithms applied to the same grouped criterion.

Table 4: Monte Carlo results, Experiment G: multiple equilibria in the data

Estimator	$\widehat{k}_1$	Bias	MCSD	$\Delta\mathscr{L}$	KL	d_∞	r_Ψ	Conv.	CPU (s)
G1: Five equilibrium assignments ($k_1^0 = 9.25$)									
1-NPL	9.2099	-0.0401	0.1502	–	0.32943	0.32787	0.4609	100	0.02
NRLS	9.2545	0.0045	0.0411	0.000	0.00028–0.00040	0.00240–0.00287	$< 10^{-6}$	100	20.7
G2: Twenty-five equilibrium assignments ($k_1^0 = 9.50$)									
1-NPL	9.5392	0.0392	0.0286	–	0.05753	0.18517	0.1870	100	0.05
1-EPL	9.1207	-0.3793	0.1600	–	0.10384	0.26187	0.1038	100	0.12
NPL	–	–	–	–	–	–	–	0	–
EPL	9.4090	-0.0910	0.0559	-543.7	0.05045	0.26430	$< 10^{-6}$	100	1.04
NRLS	9.4999	-0.0001	0.0041	0.000	4.67×10^{-6}–9.71×10^{-5}	0.00059–0.00572	$< 10^{-6}$	100	189.4

Note: $\widehat{k}_1$ is the Monte Carlo mean. $\Delta\mathscr{L}$ is the likelihood shortfall, $d_\infty = \frac{1}{2}\sum_i \|P_i - P_{i0}\|_\infty$, and $r_\Psi = \|\Psi(P;\theta) - P\|$. For estimators producing group-specific CCPs, KL, d_∞, and r_Ψ give the range of group-specific Monte Carlo averages. Conv. is the number converged out of 100; other definitions follow Table 2.

Experiment G1: Five equilibrium assignments. The grouped simultaneous-move design has $n = 5$, 55 states, $k_1 = 9.25$, and 81 MPEs at the true parameter. Five groups play equilibria $(1,1,1,55,80)$, representing three distinct equilibria. Each group contains $M_g = 5{,}000$ markets observed for $T = 5$ periods, for 125,000 market-period observations in total. NRLS recovers the assignment for every group in every replication. Its group-specific average KL divergences range from 0.00028 to 0.00040, and its maximum CCP distances range from 0.00240 to 0.00287. A single pooled 1-NPL estimate cannot represent group-specific equilibrium play and has both a large probability error and a large equilibrium residual.

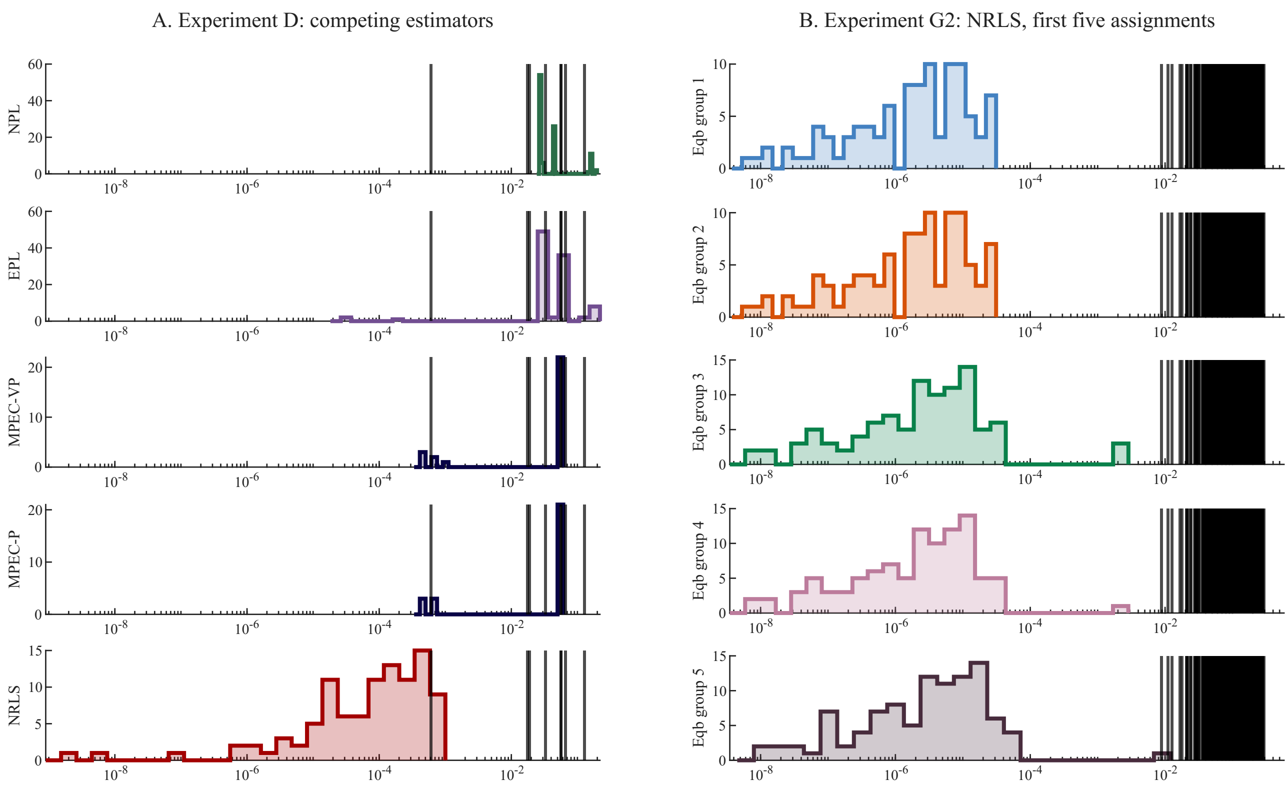
Experiment G2: Twenty-five equilibrium assignments and 19,683 MPEs. This grouped design combines extensive model multiplicity with many equilibria played in the data. The simultaneous-move game has $n = 6$, 91 states, $k_1 = 9.5$, and 19,683 MPEs at the true parameter. The 25 group assignments are $(1,1,7,7,100,\ldots,120)$, representing 23 distinct equilibria. Each group contains $M_g = 5{,}000$ markets observed for $T = 10$ periods, for 1.25 million market-period observations in total.

The computational contrast is particularly sharp. Exhaustively computing all 19,683 MPEs by RLS at a single value of the structural parameter takes approximately 2 hours and 27 minutes on the same machine used for this experiment. A brute-force full-solution estimator would have to repeat this enumeration at the trial values required by the outer optimization and derivative calculations. Complete NRLS estimation instead takes 189 seconds on average: estimating the model is substantially faster than exhaustively solving it even once.

By Proposition 3.5, this reduction costs nothing in exactness: NRLS preserves the group-specific inner maxima while never constructing all 19,683 equilibria at any trial value, and the 1.25 million observations pooled here are what make the bound discriminating enough to prune this aggressively.

NRLS identifies all 25 group assignments in all 100 replications. Its largest average group-specific KL divergence is below 10^{-4} and its largest CCP error is 0.00572. By comparison, the closest alternative MPE to a data-generating equilibrium at the true parameter has a KL divergence of approximately 10^{-2}. The pooled NPL benchmark fails to converge in every replication. Pooled EPL converges and satisfies the equilibrium restrictions for its common CCP profile, but its likelihood shortfall, parameter bias, and CCP distance show that this profile fits the heterogeneous-equilibrium data poorly. The pooled one-step estimators likewise neither impose the equilibrium restrictions nor accommodate all group-specific equilibria.

Figure 10: Distributions of equilibrium-selection accuracy



Note: The histograms report the KL divergence between the estimated and data-generating equilibria across Monte Carlo replications. Panel A compares the competing equilibrium estimators in Experiment D. Panel B reports NRLS for the first five group assignments in the design with 25 assignments and 19,683 MPEs; results for the remaining 20 assignments are similar. Vertical lines indicate the KL distances from each data-generating equilibrium to the alternative MPEs at the true parameter.

Figure 10 complements the averages in Tables 2 and 4. Panel A shows that the equilibrium-selection errors in Experiment D are not driven by a few innocuous outliers: the local estimators repeatedly select policies separated from the data-generating equilibrium by economically meaningful KL distances. Panel B reports NRLS for the first five assignments in Experiment G2. Its errors are concentrated orders of magnitude below the distances to the alternative MPEs. This is the finite-sample empirical analogue of the population separation condition in Assumption B(ii): pooling many markets within each group makes the data-generating assignments distinguishable despite the presence of 19,683 equilibria.

5 Discussion and conclusions

Multiplicity in dynamic games is bad news for estimation methods that rely on a local solution. Our experiments show why: NPL can fail to converge, EPL can converge to a locally coherent but globally inferior branch, and local MPEC solvers can satisfy the equilibrium constraints while missing the likelihood-maximizing equilibrium — failures that grow more severe as the equilibrium set expands and near discontinuities of the full-solution likelihood, where the maximizing equilibrium changes abruptly with the parameter. A small equilibrium residual or successful numerical convergence is therefore not enough to establish that a structural estimate is globally credible. Results from local methods should be subjected to serious falsification attempts whenever multiplicity is plausible.

NRLS provides a constructive full-solution benchmark. Our one-step definition of directionality — equivalent to common block-triangularity of the transition matrices, and checkable directly from the model primitives — delimits the class; within it, RLS represents every Markov perfect equilibrium as a root-to-leaf path in a recursively discovered tree, and the nonparametric-likelihood bound lets branch and bound prune that tree without sacrificing the discrete optimum. Conditional on finding all stage equilibria, NRLS therefore computes the exact full-solution MLE while solving only the stage games on the part of the tree that the data cannot rule out. When observations pooled under a common equilibrium increase, the bounds separate inferior branches earlier, so the computational burden can fall rather than rise with sample size.

The Monte Carlo experiments provide the finite-sample counterpart of the asymptotic theory: NRLS remains reliable in designs with unstable equilibria, likelihood cliffs, and thousands of candidate equilibria, recovers heterogeneous equilibrium assignments across known groups of markets, and, in the largest design, estimates the model substantially faster than the time it takes to exhaustively solve it even once.

Important qualifications remain. Exactness applies to the discrete maximization over the equilibrium set produced by the stage-game solver, and requires that solver to find every stage equilibrium: exact and non-iterative for the leapfrogging stage games when the private-shock scale is zero, a piecewise-linear approximation of best responses at positive scale. The computational gains also exploit directionality and do not automatically extend to unrestricted dynamic games. Our multiple-equilibrium experiments condition on a known partition of markets into groups and do not estimate latent group membership; clustering methods such as Bonhomme and Manresa (2015) and Bonhomme, Lamadon and Manresa (2022) provide a natural next step.

Inference is especially delicate when the true parameter lies near the boundary of the equilibrium-specific set $\Theta(e_0)$. The oracle normal approximation in Section 3.1 assumes that θ_0 is in the interior of this set; at the boundary the limiting distribution is nonstandard (Andrews, 1999). The simulation-based procedure in Appendix A.4 is a starting point for quantifying small-sample uncertainty, but local-to-boundary asymptotics and practical confidence procedures remain open.

Finally, reliable counterfactual analysis under multiplicity requires more than agreement on structural parameters. Our experiments show that methods can deliver similar parameter estimates while selecting equilibria with very different choice probabilities, and those equilibria can imply sharply different counterfactuals. Future work should therefore combine full-solution estimation with procedures that trace the set of counterfactual equilibria through bifurcations—for example, path-following or homotopy methods anchored at the estimated baseline equilibrium—rather than reporting a counterfactual from an arbitrary local branch. Together with latent-group estimation and boundary-robust inference, this is the main route from an exact likelihood benchmark to credible empirical use of dynamic games with multiple equilibria.

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A Proofs

A.1 Proof of Proposition 2.1

Throughout we use the notation of Section 2.2 and maintain assumptions (AS), (CI) and (IPV) of Section 2.1; assumption (EV) is not needed, and absolute continuity of the distributions of the private shocks is sufficient.

Proof of Proposition 2.1. First observe that for any $\sigma \in \Sigma$ and any total order $\trianglelefteq$ on D, the block-triangularity condition (8) states precisely that every edge $d \to d'$ of the graph $\mathscr{D}(\sigma)$ satisfies $d \triangleleft d'$, i.e. that $\trianglelefteq$ is a *topological order* of $\mathscr{D}(\sigma)$.

(Only if.) Let the game be directional with respect to $X = D \times X'$. By condition (1) of Definition 3 applied to a completely mixed profile $\bar{\sigma} \in \Sigma$, the maximal transition graph $\bar{\mathscr{D}} = \mathscr{D}(\bar{\sigma})$ is acyclic, and being finite it admits a topological order. Indeed, a finite acyclic graph must contain a vertex with no outgoing edges, since otherwise, starting anywhere and repeatedly following outgoing edges, one would eventually revisit a vertex, closing a cycle; place such a vertex last in the order, remove it from the graph, and repeat on the remaining graph, which is again acyclic, filling the positions of the order from the end. Every edge $d \to d'$ then satisfies $d \triangleleft d'$, because at the step when d was removed it had no outgoing edges among the remaining vertices, so d' had been removed earlier and received a later position. By the observation above, $F^{\bar{\sigma}}$ is upper block-triangular with respect to the constructed order $\trianglelefteq$; and since every $\mathscr{D}(\sigma)$, $\sigma \in \Sigma$, is a subgraph of $\bar{\mathscr{D}}$ (Section 2.2), every edge of every $\mathscr{D}(\sigma)$ likewise satisfies $d \triangleleft d'$, so F^{σ} is upper block-triangular with respect to the same common order for every $\sigma \in \Sigma$.

(If.) Conversely, let $\trianglelefteq$ be a total order on D such that $F^{\bar{\sigma}}$ is upper block-triangular for some completely mixed $\bar{\sigma}$ and thus, by the maximality of the set of non-zero elements of $F^{\bar{\sigma}}$ established in Section 2.2, for every $\sigma \in \Sigma$. By the observation above, $\trianglelefteq$ is a topological order of every graph $\mathscr{D}(\sigma)$, and a graph that admits a topological order cannot contain a directed cycle, because following the edges of a cycle $d_1 \to \cdots \to d_k \to d_1$ would yield $d_1 \triangleleft d_2 \triangleleft \cdots \triangleleft d_k \triangleleft d_1$, contradicting that $\trianglelefteq$ is an order. Hence condition (1) of Definition 3 holds for every $\sigma \in \Sigma$, and condition (2) follows since it is implied by condition (1), as noted in Section 2.2. $\square$

It remains to relate the directionality conditions (1) and (2) of Definition 3, which are formulated in terms of *one-step* transitions, to the original definitions in IRS2016, which are formulated in terms of *hitting probabilities*. For $\sigma \in \Sigma$, $d \neq d'$ and $x \in B_d$, let $\rho(d'|d,x,\sigma)$ denote the probability that the Markov chain $\{x_t\}$ with transition matrix F^{σ} started at $x_0 = x$ visits the substage $B_{d'}$ at some period $t \geq 1$ (IRS2016, footnote 9). Since the state space is finite, $\rho(d'|d,x,\sigma) > 0$ if and only if there is a finite sequence of states $x = x_0, x_1, \ldots, x_n$, $n \geq 1$, with $F^{\sigma}_{x_{k-1},x_k} > 0$ for $k = 1, \ldots, n$ and $x_n \in B_{d'}$; we refer to such a sequence as a *positive probability path* from x to $B_{d'}$ under σ. To shorten the notation, we say that *d leads to d' under σ*, denoted $d \rightsquigarrow_{\sigma} d'$, if $\rho(d'|d,x,\sigma) > 0$ for at least one $x \in B_d$. In this notation, Definition 2 of IRS2016 introduces the strategy-specific binary relation $\succ_{\sigma}$ on D by declaring $d' \succ_{\sigma} d$ if and only if $d \rightsquigarrow_{\sigma} d'$ and not $d' \rightsquigarrow_{\sigma} d$; their *no loop* condition (Definition 3) requires every $\succ_{\sigma}$-non-comparable pair of distinct d, d' not to communicate, which is equivalent to requiring that no pair of distinct d, d' satisfies both $d \rightsquigarrow_{\sigma} d'$ and $d' \rightsquigarrow_{\sigma} d$ (a pair with one-way communication is comparable by the definition of $\succ_{\sigma}$, so a non-comparable pair either does not communicate or communicates both ways); their *consistency* condition (Definition 4) requires that no pair is ordered oppositely, $d' \succ_{\sigma} d$ and $d \succ_{\sigma'} d'$, by two profiles $\sigma, \sigma' \in \Sigma$; and the game is a *DDG* with respect to $X = D \times X'$ (Definition 5) if every $\sigma \in \Sigma$ has no loops and the relations $\{\succ_{\sigma} : \sigma \in \Sigma\}$ are pairwise consistent.

Lemma A.1 (Directionality implies the DDG conditions of IRS2016)**.** *Let the game be directional with respect to $X = D \times X'$ in the sense of Definition 3, and let $\trianglelefteq$ be a total order on D for which the block-triangularity in Proposition 2.1 holds. Then:*

(i) the game is a DDG with respect to $X = D \times X'$ in the sense of Definition 5 of IRS2016;

(ii) every strategy-specific relation satisfies $\succ_\sigma \subseteq \rhd$, where $d' \rhd d$ denotes $d \lhd d'$, and consequently the coarsest common refinement $\succ^* = TC\big(\cup_{\sigma\in\Sigma} \succ_\sigma\big)$, where $TC(\cdot)$ denotes the transitive closure, satisfies $\succ^* \subseteq \rhd$ and is a strict partial order on D. Thus the conclusion of Theorem 1 of IRS2016 (existence of the strategy-independent partial order, denoted $\succ_G$ there, with G standing for the game) holds, and with it the constructions and results of IRS2016 that build on it, including the stages of the game, the state recursion algorithm and the recursive lexicographical search algorithm.

Proof. Fix any $\sigma \in \Sigma$. Along any positive probability path under σ the substage index can only move up the order $\unlhd$: for every step $F^\sigma_{x_{k-1},x_k} > 0$ with $x_{k-1} \in B_d$, $x_k \in B_{d'}$, either $d = d'$ or $d \to d'$ is an edge of $\mathscr{D}(\sigma)$ and thus $d \lhd d'$ by (8). Consequently,

$$d \rightsquigarrow_\sigma d' \text{ and } d \neq d' \quad \Longrightarrow \quad d \lhd d'. \tag{31}$$

(i) The two conditions of Definition 5 of IRS2016 follow from (31). No loops: if a pair of distinct d, d' communicated both ways under some σ, i.e. $d \rightsquigarrow_\sigma d'$ and $d' \rightsquigarrow_\sigma d$, then both $d \lhd d'$ and $d' \lhd d$ would hold, which is impossible. Consistency: if $d' \succ_\sigma d$ for some σ, then $d \rightsquigarrow_\sigma d'$ and thus $d \lhd d'$; if in addition $d \succ_{\sigma'} d'$ for some σ', then $d' \rightsquigarrow_{\sigma'} d$ and thus $d' \lhd d$, again a contradiction.

(ii) If $d' \succ_\sigma d$ then $d \rightsquigarrow_\sigma d'$ with $d \neq d'$, so $d \lhd d'$ by (31), i.e. $\succ_\sigma \subseteq \rhd$. The relation $\rhd$ is transitive, so it contains the transitive closure of any of its subsets; hence $\succ^* = TC\big(\cup_\sigma \succ_\sigma\big) \subseteq \rhd$. As a subset of the strict total order $\rhd$, the relation $\succ^*$ is irreflexive and asymmetric, and it is transitive by construction, so it is a strict partial order on D that refines every $\succ_\sigma$. It is also the *coarsest* common refinement: any common refinement of $\{\succ_\sigma : \sigma \in \Sigma\}$ is a transitive relation containing $\cup_\sigma \succ_\sigma$, and therefore contains its transitive closure $\succ^*$. Finally, (31) holds for every feasible profile and states that under any $\sigma \in \Sigma$ play never returns to a $\unlhd$-earlier substage; this is precisely the stage monotonicity property that the decomposition and subgame perfection results of IRS2016 rely upon, in addition to the existence of the strategy-independent partial order, so the constructions built upon these properties apply. $\square$

We note that the converse of Lemma A.1(i) fails, so the one-step conditions (1) and (2) are strictly stronger than the hitting-probability-based conditions of IRS2016, and the strengthening is substantive. The conditions of IRS2016 constrain pairs of substages only through the *existence* of positive probability paths, and when the diagonal blocks of $F^{\bar\sigma}$ are not irreducible, the state at which a substage is entered may differ from the states at which the paths continuing to a third substage originate, so that one-step cycles through three or more substages need not produce two-way communication between any pair. As an illustration, let $D = \{1,2,3\}$, $X' = \{1,2\}$, let the transition probabilities be action-independent with $(1,1) \to (2,2)$, $(2,1) \to (3,2)$, $(3,1) \to (1,2)$ and all states $(d,2)$ absorbing. The maximal transition graph contains the cycle $1 \to 2 \to 3 \to 1$, so the game is not directional with respect to $D \times X'$ in the sense of Definition 3 and F^σ admits no block-triangular form with respect to these substages; and yet each substage leads only to the next one along the cycle, so no two distinct substages communicate both ways and the conditions of Definition 5 of IRS2016, read literally, hold (the finer decomposition of this example with singleton X' is, of course, directional). Moreover, in this game every $\sigma \in \Sigma$ induces the relations $2 \succ_\sigma 1$, $3 \succ_\sigma 2$ and $1 \succ_\sigma 3$, so the transitive closure $\succ^* = TC\big(\cup_{\sigma\in\Sigma} \succ_\sigma\big)$ contains both $3 \succ^* 1$ and $1 \succ^* 3$ and fails to be asymmetric: contrary to the conclusion of Theorem 1 of IRS2016, no strategy-independent partial order refining the relations $\succ_\sigma$ exists in this game. Definition 5 of IRS2016, read literally, thus admits games for which the conclusion of their Theorem 1, and with it the constructions that build on the strategy-independent partial order, fail. The one-step conditions (1) and (2) close this gap: accordingly, Lemma A.1 derives the strategy-independent partial order and the stage monotonicity of play directly from the one-step conditions, without relying on the derivation in IRS2016. For these reasons we treat the one-step conditions (1) and (2), which are also the ones verifiable directly from the model primitives, as the operative definition of directionality, and record for completeness the standard cases in which the two formulations coincide. If $|X'| = 1$, every substage is a single state, so positive probability paths concatenate across substages, and any directed cycle in $\bar{\mathscr{D}}$ yields two-way communication between a pair of its vertices, violating the no loop condition for $\bar{\sigma}$. The same argument applies whenever each diagonal block of $F^{\bar{\sigma}}$ is irreducible (every substage communicates internally), since a positive probability path can then be extended within the substage from its entry state to the origin of the next edge of the cycle.

A.2 Proof of Theorem 3.1

Proof. First note that Assumption B(i), (iii) and (iv) imply $\sup_{\theta\in\Theta(e)} |\mathscr{L}_{MT}(\theta,e) - \mathscr{L}_T(\theta,e)| \to^P 0$ for all $e \in \mathscr{E}$, c.f. Lemma 2.1 in Newey and McFadden (1994). This in turn implies that

$$\sup_{e\in\mathscr{E}} \sup_{\theta\in\Theta(e)} |\mathscr{L}_{MT}(\theta,e) - \mathscr{L}_T(\theta,e)| \to^P 0 \tag{32}$$

since $\mathscr{E}$ is a finite set. Second, Assumption B(i)-(iii) imply that, for any given $\varepsilon > 0$, there exists a $\delta > 0$ such that $\|\theta - \theta_0\| > \varepsilon$ and $e \neq e_0$ implies $\mathscr{L}_T(\theta,e) \leq \mathscr{L}_T(\theta_0,e_0) + \delta$. Note this in particular holds true if $\theta \notin \Theta(e)$ since in this case $\mathscr{L}_T(\theta,e) = -\infty$.

We wish to show that for any $\varepsilon > 0$, $P\left(\left\{||\hat{\theta} - \theta_0|| > \varepsilon\right\} \bigcup \{\hat{e} \neq e_0\}\right) \to 0$ as $M \to \infty$. As shown above, for a given $\varepsilon > 0$, there exists a $\delta > 0$ such that $\|\theta - \theta_0\| > \varepsilon$ or $e \neq e_0$ implies $\mathscr{L}_T(\theta,e) \leq \mathscr{L}_T(\theta_0,e_0) + \delta$. This in turn implies $|\mathscr{L}_T(\theta,e) - \mathscr{L}_T(\theta_0,e_0)| \geq \delta$. Thus,

$$\Pr(||\hat{\theta} - \theta_0|| > \varepsilon, \hat{e} \neq e_0) \leq \Pr(\mathscr{L}_T(\hat{\theta},\hat{e}) \leq \mathscr{L}_T(\theta_0,e_0) + \delta) \leq \Pr(|\mathscr{L}_T(\hat{\theta},\hat{e}) - \mathscr{L}_T(\theta_0,e_0)| \geq \delta).$$

We then wish to show that the final probability in above display converges to zero which is equivalent to $\mathscr{L}_T(\hat{\theta},\hat{e}) \to^P \mathscr{L}_T(\theta_0,e_0)$. Since (θ_0,e_0) is the unique maximizer of $\mathscr{L}_T(\theta,e)$, we know that $\mathscr{L}_T(\theta_0,e_0) \geq \mathscr{L}_T(\hat{\theta},\hat{e})$. Thus,

$$|\mathscr{L}_T(\hat{\theta},\hat{e}) - \mathscr{L}_T(\theta_0,e_0)| = \mathscr{L}_T(\theta_0,e_0) - \mathscr{L}_T(\hat{\theta},\hat{e}) = \left\{\mathscr{L}_T(\theta_0,e_0) - \hat{\mathscr{L}}_T(\theta_0,e_0)\right\} + \left\{\hat{\mathscr{L}}_T(\theta_0,e_0) - \mathscr{L}_T(\hat{\theta},\hat{e})\right\}$$

where, by (32), $\mathscr{L}_T(\theta_0,e_0) - \hat{\mathscr{L}}_T(\theta_0,e_0) = o_P(1)$. For the second right–hand side term of above display, first observe that, by the definition of $(\hat{\theta},\hat{e})$, $\hat{\theta} \in \Theta(\hat{e})$ for some $\hat{e} \in \mathscr{E}$. Thus, again using (32),

$$\hat{\mathscr{L}}_T(\theta_0,e_0) - \mathscr{L}_T(\hat{\theta},\hat{e}) \leq \hat{\mathscr{L}}_T(\hat{\theta},\hat{e}) - \mathscr{L}_T(\hat{\theta},\hat{e}) \leq \max_{e\in\mathscr{E}} \sup_{\theta\in\Theta(e)} \left|\hat{\mathscr{L}}_T(\theta,e) - \mathscr{L}_T(\theta,e)\right| \to^P 0.$$

In conclusion, $\mathscr{L}_T(\hat{\theta},\hat{e}) \to^P \mathscr{L}_T(\theta_0,e_0)$ as desired. □

A.3 Proof of Theorem 3.2

Proof. As part of the proof of Theorem 3.1, it was shown that $P(||\hat{\theta} - \theta_0|| < \varepsilon, \hat{e} = e_0) \to 1$ as $M \to \infty$. In particular, $\hat{\theta} = \hat{\theta}_0$ with probability approaching one (w.p.a.1). Now, using Assumption 2(i)-(ii) and the mean–value theorem,

$$0 = S_{M,T}\left(\hat{\theta},\hat{e}\right) = S_{M,T}\left(\hat{\theta},e_0\right) = S_{M,T}(\theta_0,e_0) + H_{M,T}\left(\bar{\theta},e_0\right)(\hat{\theta} - \theta_0),$$

w.p.a.1, for some $\bar{\theta}$ lying on the line segment connecting $\hat{\theta}$ and θ_0, where

$$S_{M,T}(\theta,e) = \frac{1}{MT}\sum_{m=1}^{M}\sum_{t=1}^{T} s(z_{m,t_1};\theta,e), \quad H_{M,T}(\theta,e) = \frac{1}{MT}\sum_{m=1}^{M}\sum_{t=1}^{T} h(z_{m,t_1};\theta,e). \tag{33}$$

By Assumption 2 (iii)-(iv) together with the CLT and the ULLN, $\sqrt{M}S_{M,T}(\theta_0, e_0) \to^d N(0, \Omega_{T,0})$ and $H_{M,T}(\bar{\theta}, e_0) \to^p H_{T,0}$. The result now follows by Slutsky's Theorem. □

A.4 Inference on equilibrium

Proof of Proposition 3.3. Step 1. Fix $e \in \mathscr{E}$. Since the closure of $\Theta(e) \subseteq \Theta$ is compact by Assumption B(i), and $\theta_0(e)$ is its unique maximizer of $\mathscr{L}_T(\cdot, e)$ by Assumption D(i), the uniform law of large numbers used in the proof of Theorem 3.1, cf. (32), yields consistency of the equilibrium–specific estimator, $\hat{\theta}(e) \to^P \theta_0(e)$, by the standard argument for extremum estimators. Since $\theta_0(e) \in \operatorname{int}\Theta(e)$, $\hat{\theta}(e)$ is an interior maximizer w.p.a.1, and under Assumption D(ii) the proof of Theorem 3.2 applies verbatim at $(\theta_0(e), e)$ in place of (θ_0, e_0), giving

$$\sqrt{M}(\hat{\theta}(e) - \theta_0(e)) \to^d N(0, V_T(e)), \quad V_T(e) = H_{T,0}^{-1}(e)\,\Omega_{T,0}(e)\,H_{T,0}^{-1}(e), \tag{34}$$

where $H_{T,0}(e) = H_T(\theta_0(e), e)$ and $\Omega_{T,0}(e) = \sum_{t_1,t_2=1}^{T} E\left[s(z_{m,t_1}; \theta_0(e), e)\, s(z_{m,t_2}; \theta_0(e), e)'\right]/T^2$.

Step 2. Expanding $\mathscr{L}_{MT}(\cdot, e)$ around $\hat{\theta}(e)$, where the score vanishes w.p.a.1 by interiority, the mean–value theorem gives

$$M\left\{\mathscr{L}_{MT}(\hat{\theta}(e), e) - \mathscr{L}_{MT}(\theta_0(e), e)\right\} = -\frac{M}{2}(\hat{\theta}(e) - \theta_0(e))' H_{M,T}(\bar{\theta}(e), e)(\hat{\theta}(e) - \theta_0(e)) = O_P(1),$$

for some intermediate $\bar{\theta}(e)$, by (34) and $H_{M,T}(\bar{\theta}(e), e) \to^P H_{T,0}(e)$, with $S_{M,T}$ and $H_{M,T}$ defined in (33). Hence $\sqrt{M}\left\{\mathscr{L}_{MT}(\hat{\theta}(e), e) - \mathscr{L}_{MT}(\theta_0(e), e)\right\} = O_P(M^{-1/2}) = o_P(1)$.

Step 3. Let $W_m = \left(\frac{1}{T}\sum_{t=1}^{T} \ell(z_{mt}; \theta_0(e), e)\right)_{e \in \mathscr{E}} \in R^{|\mathscr{E}|}$. The vectors W_m are i.i.d. across markets with $E\left[\|W_m\|^2\right] < \infty$ by Assumption B(iv) and Assumption D(iii), so the multivariate central limit theorem gives, jointly across $e \in \mathscr{E}$,

$$\sqrt{M}\left(\mathscr{L}_{MT}(\theta_0(e), e) - \mathscr{L}_T(\theta_0(e), e)\right)_{e \in \mathscr{E}} \to^d Z \sim N(0, \Sigma_T),$$

with Σ_T as defined in Proposition 3.3(i). Adding the $o_P(1)$ terms of Step 2 coordinate by coordinate ($\mathscr{E}$ is finite) yields $Z_M \to^d Z$ jointly, which is (17) and proves part (i).

Step 4. For part (ii), fix $e \in \mathscr{E}$ and write $\Delta_M(e, e') = \sqrt{M}\{\mathscr{L}_T(\theta_0(e'), e') - \mathscr{L}_T(\theta_0(e), e)\}$, so that

$$\{\hat{e} = e\} = \bigcap_{e' \neq e}\left\{Z_M(e) - Z_M(e') \geq \Delta_M(e, e')\right\}, \quad \{\hat{e}_M^* = e\} = \bigcap_{e' \neq e}\left\{Z(e) - Z(e') \geq \Delta_M(e, e')\right\},$$

up to ties, which occur with probability zero in the limit problem by the nondegeneracy condition of part (ii) and with probability tending to zero in the sample problem, and are therefore immaterial under any tie–breaking rule. Partition $\mathscr{E} \setminus \{e\}$ according to the sign of $\mathscr{L}_T(\theta_0(e'), e') - \mathscr{L}_T(\theta_0(e), e)$. If some e' has $\mathscr{L}_T(\theta_0(e'), e') > \mathscr{L}_T(\theta_0(e), e)$ then $\Delta_M(e, e') \to +\infty$ while $Z_M(e) - Z_M(e') = O_P(1)$ and $Z(e) - Z(e') = O_P(1)$, so both probabilities in (18) converge to zero and the claim holds. Otherwise, for every e' with $\mathscr{L}_T(\theta_0(e'), e') < \mathscr{L}_T(\theta_0(e), e)$ the corresponding constraint holds w.p.a.1 in both problems, since $\Delta_M(e, e') \to -\infty$; and for every e' in the set $D_0(e)$ of equilibria with $\mathscr{L}_T(\theta_0(e'), e') = \mathscr{L}_T(\theta_0(e), e)$ the constraint reads $Z_M(e) - Z_M(e') \geq 0$ and $Z(e) - Z(e') \geq 0$, respectively. Therefore both probabilities in (18) differ by $o(1)$ from the probability of $\bigcap_{e' \in D_0(e)}\{Z_M(e) - Z_M(e') \geq 0\}$ and its analogue with Z, and these converge to each other by part (i) and the continuous mapping theorem, the boundary of the limit event having probability zero because each difference $Z(e) - Z(e')$, $e' \in D_0(e)$, is a nondegenerate normal by the condition of part (ii). □